

Supplementary Material

Cooperation Without Adaptation: LLM Agents in a Collective-Risk Dilemma

Anonymous submission

This document supports the paper. The paper states its results in words and keeps its numbers in tables and figures; this supplement gives the values behind every statement, the full proofs, every prompt and question word for word, and the methods. Section numbers here start with S; “Section 4” without an S means a section of the paper.

Contents

S1	How to read this supplement	3
S2	The game and its benchmarks	3
S2.1	Rules and notation	3
S2.2	Proposition 1 and its proof	3
S2.3	The ten-round game	4
S2.4	Proposition 2 and its proof	5
S2.5	The optimum and the three ways to lose	5
S2.6	How much risk aversion would explain paying at low risk	5
S2.7	Scripted partners and their best responses	6
S3	Models and protocol	6
S3.1	Models	7
S3.2	How a round is played	7
S3.3	How replies are read	7
S4	Prompts, word for word	7
S4.1	Base prompt	7
S4.2	No hint	8
S4.3	Neutral wording	8
S4.4	Neutral wording, goal stated	9
S4.5	Two rewordings	9
S5	Designs, data and statistics	11
S5.1	Games in each design	11
S5.2	The lottery and why the paper uses expected payoffs	11
S5.3	Statistical methods	12
S5.4	Reproducibility	13
S6	Self-play across risk	13
S6.1	Numbers behind the main text	13
S6.2	What a seat pays at each risk level	13
S6.3	Outcomes at each risk level	15
S6.4	Where the payoff goes	15
S6.5	The end of the game	16
S7	Prompt and decoding controls	16
S7.1	Numbers behind the main text	17
S7.2	Every change against the base grid	17
S7.3	The goal test at each risk level	19
S7.4	How each model’s signature changes	19

S7.5	Removing the hint	20
S8	In-game questions	22
S8.1	Numbers behind the main text	22
S8.2	A question as the agent receives it	22
S8.3	The ten questions	23
S8.4	Rule questions	24
S8.5	The value comparison	24
S8.6	Answers against play	25
S8.7	The second value question	27
S9	One model beside scripted partners	27
S9.1	Numbers behind the main text	28
S9.2	What the models pay	28
S9.3	Outcomes and distance from the best response	30
S9.4	Partner versus risk	31
S9.5	Paying after the outcome is settled	32
S9.6	Fair-share partners	32
S10	Two models at one table	33
S10.1	Numbers behind the main text	33
S10.2	Who earns more at a shared table	33
S10.3	Which tables reach the target	34
S10.4	The last round	35
S10.5	Slack and generous partners	36
S11	Selection among models	38
S11.1	Method	38
S11.2	Numbers behind the main text	39
S11.3	Payoff functions	39
S11.4	Every selection setting	42
S11.5	Facts about single tables	45
S11.6	Single entrants, candidate sets and welfare	46
S12	Strategies, the last round and equilibrium benchmarks	48
S12.1	The last round	48
S12.2	Play against the equilibria of Proposition 1	48
S12.3	Simple rules describe the moves	49
S12.4	Responses to Grok 4.20's opening move	50

S1 How to read this supplement

Table S1 maps each part of the paper to the sections here. Every results section (Sections S6 to S11) opens with a short table called “Numbers behind the main text”, which lists each statement the paper makes in words, stated precisely, beside its value. The tables and figures that follow give the full breakdown by model, risk level and condition.

Table S1: Where to find what.

Part of the paper	In this supplement
Section 3.1, rules and Propositions 1 and 2	Section S2: full proofs, the ten-round game, the loss decomposition, risk aversion, best responses to scripted partners
Section 3.2, models, prompts, protocol	Sections S3 and S4: model identifiers, how replies are read, every prompt word for word
Sections 3.2 and 3.3, designs and measures	Section S5: games per design, the lottery draws, statistical methods, reproducibility
Section 4.1.1, self-play across risk	Section S6
Section 4.1.2, prompt and decoding changes	Section S7
Section 3.3, asking the agent	Section S8: every question word for word, all answers, and answers against play
Section 4.2.1, fixed partners	Section S9
Section 4.2.2, mixed tables	Section S10
Sections 3.3 and 4.3, selection	Section S11: method, payoff functions, all selection settings
Sections 4.1.1 and 4.2.2, last round, equilibria, rules, responses to openings	Section S12: including how each model plays the last round

Conventions. The unit of analysis is the game, because the seats of one game share a history and a lottery. Payoffs are expected payoffs over the catastrophe lottery (Section S5.2). A seat’s *total* is what it paid over the ten rounds, out of 40. Intervals are 95% percentile bootstrap intervals over games, written [lower, upper]. Models always appear in the same order: Claude Haiku 4.5, Gemini 3.5 Flash-Lite, GPT-5.6 Luna, Qwen3-235B, Grok 4.20.

S2 The game and its benchmarks

S2.1 Rules and notation

Six players each start with an endowment $e = 40$. In each of $R = 10$ rounds, every player chooses privately and at the same time a contribution $a_i^t \in \{0, 2, 4\}$ to a shared pool, and after each round every seat’s contribution is shown to all. A player’s total is $x_i = \sum_{t=1}^R a_i^t \in \{0, 2, \dots, 40\}$ and the pool is $X = \sum_{i=1}^6 x_i$. If $X \geq T = 120$, every player keeps $e - x_i$. If $X < T$, a single lottery for the whole table destroys all savings with probability p , and otherwise every player keeps $e - x_i$. For a risk-neutral player the expected payoff is therefore

$$u_i(x) = \begin{cases} e - x_i & \text{if } X \geq T, \\ (1 - p)(e - x_i) & \text{if } X < T. \end{cases} \quad (1)$$

The fair share is $T/6 = 20$ per player, or 2 per round, and the pivot is $p^* = T/(6e) = \frac{1}{2}$. The equilibrium results below assume $p < 1$. At $p = 1$ a missed target leaves every player with nothing whatever it paid, so many profiles that miss the target become weak equilibria and the characterisation changes.

S2.2 Proposition 1 and its proof

Proposition 1. *Consider the game in which each player picks a total $x_i \in \{0, 2, \dots, 40\}$, with payoffs (1) and $p < 1$. A profile is a Nash equilibrium if and only if $x = 0$, or $X = T$ and $x_i \leq pe$ for every i . A target-reaching equilibrium exists if and only if $p \geq p^*$. The welfare-optimal outcome is $x = 0$ for $p < p^*$ and $X = T$ for $p > p^*$, and both are optimal at $p = p^*$, so the best attainable payoff per player is $\text{opt}(p) = \max\{(1 - p)e, e - T/6\}$.*

Proof. We check the three cases $X < T$, $X > T$ and $X = T$ in turn.

Case $X < T$. Suppose some player has $x_i > 0$. If it pays nothing instead, the pool is still below T and its payoff rises from $(1 - p)(e - x_i)$ to $(1 - p)e$, which is strictly more because $p < 1$. So a profile that misses the target can be an equilibrium only

if $x = 0$. Conversely, $x = 0$ is an equilibrium: a single player can raise the pool to at most $40 < T$, so any deviation $x'_i > 0$ yields $(1-p)(e-x'_i) < (1-p)e$.

Case $X > T$. Totals are even and T is even, so $X \geq T + 2$. Some player has $x_i > 0$ and can pay 2 less. The pool stays at or above T , and its payoff rises by 2. No such profile is an equilibrium.

Case $X = T$. Paying more lowers a player's payoff, since the target is met anyway. Paying less makes the pool miss the target, and the best such deviation is to pay nothing, which yields $(1-p)e$. So player i has no profitable deviation if and only if $e - x_i \geq (1-p)e$, that is $x_i \leq pe$.

Existence. If $p \geq p^*$, then $pe \geq 20$ and the profile in which every player pays 20 is an equilibrium. If $p < p^*$, then $pe < 20$, every equilibrium total is below 20, and six of them cannot add up to $T = 120$.

Welfare. Total welfare is $6e - X$ when $X \geq T$ and $(1-p)(6e - X)$ when $X < T$. Over profiles that reach the target it is largest at $X = T$, with value $6e - T$; over profiles that miss it, it is largest at $X = 0$, with value $(1-p)6e$. The first is larger if and only if $p > T/(6e) = p^*$, and they are equal at p^* . Dividing by six gives $\text{opt}(p)$. $\square$

The welfare statement also holds at $p = 1$, as the paper states: a missed target then leaves welfare 0, below the $6e - T$ of a pool that reaches T exactly, so $\text{opt}(1) = e - T/6$, which is what the formula gives.

The equilibrium statement is also checked against every unilateral deviation, from every symmetric profile and from 3,000 random asymmetric profiles at each of the ten risk levels from 0 to 0.9, as part of the tests that will be released with the final version; the case $p = 1$ is checked separately.

S2.3 The ten-round game

In the ten-round game a pure strategy maps every history of past contributions to a contribution in $\{0, 2, 4\}$, and payoffs still depend only on the totals through (1).

Proposition S1. *Let $p < 1$. The totals x reached on the path of a pure-strategy Nash equilibrium of the ten-round game are exactly the equilibrium profiles of Proposition 1. Each of them is reached by an equilibrium in which every player follows a fixed schedule that ignores the history.*

Proof. Every profile of Proposition 1 is reached. Let each player follow any fixed schedule of contributions in $\{0, 2, 4\}$ that adds up to x_i ; one exists because $x_i \leq 40$. The other players' totals then do not depend on what player i does, so whatever strategy i uses, it ends with some total x'_i against fixed totals of the others. Its payoff is the payoff of the one-shot totals game, which by Proposition 1 is at most the payoff of x_i . No deviation is profitable.

Only such profiles are reached. Fix a pure equilibrium and its path totals x .

(i) The strategy "always pay 0" guarantees at least $(1-p)e$ against anything, since it earns e if the target is met and $(1-p)e$ otherwise. In equilibrium every player therefore earns at least $(1-p)e$.

(ii) If $X < T$, a player earns $(1-p)(e - x_i)$, which is at least $(1-p)e$ only if $x_i = 0$. So $x = 0$.

(iii) If $X \geq T$, a player earns $e - x_i \geq (1-p)e$, so $x_i \leq pe$.

(iv) It remains to rule out $X > T$. Let τ be the first round after which the path pool is at least T . If some player i pays a positive amount in a round after τ , let t' be the last such round for that player, and let i follow its strategy up to round $t' - 1$ and pay 0 from round t' on. The target is already met after round τ and contributions cannot be negative, so it stays met whatever the others do later, and i 's total falls by at least 2: a profitable deviation. Otherwise nobody pays after τ , so the pool after round τ equals $X \geq T + 2$. Some player i pays $a_i^\tau \geq 2$ in round τ . Let i pay $a_i^\tau - 2$ in round τ and 0 afterwards. Decisions within a round are simultaneous, so the others' round- τ contributions are unchanged, the pool after round τ is $X - 2 \geq T$, and i gains 2. Either way the profile is not an equilibrium, so $X = T$.

Together, (ii) to (iv) give exactly the conditions of Proposition 1. $\square$

We claim nothing about subgame perfection: the fixed schedules above need not be optimal after histories that are off the path.

S2.4 Proposition 2 and its proof

Proposition 2. For two seats i and j in the same game, $u_i - u_j = c(x_j - x_i)$, with $c = 1$ if the target was reached and $c = 1 - p$ otherwise.

Proof. Both seats face the same pool X . If $X \geq T$, $u_i - u_j = (e - x_i) - (e - x_j) = x_j - x_i$. If $X < T$, $u_i - u_j = (1 - p)((e - x_i) - (e - x_j)) = (1 - p)(x_j - x_i)$. $\square$

The lottery is drawn once per table, so the same ordering holds for realised payoffs: if the catastrophe strikes, both seats get 0, and otherwise the seat that paid less keeps more. The analysis code checks the identity against the game engine's own accounts in all 1,500 mixed games, with a largest error below 10^{-12} .

S2.5 The optimum and the three ways to lose

For a game at risk p , a model's loss is $\text{opt}(p) - \bar{u}$, where $\bar{u}$ is the mean expected payoff of its seats. In self-play all six seats hold the same model, so $\bar{u} = e - X/6$ if the target is reached and $(1 - p)(e - X/6)$ otherwise. The paper splits the loss of each game into three non-negative parts that add up to it:

$$\begin{aligned} \text{paying past } 120 &= \begin{cases} (X - T)/6 & \text{if } X \geq T, \\ 0 & \text{otherwise;} \end{cases} \\ \text{paying below } p^* &= \begin{cases} \text{opt}(p) - (e - T/6) & \text{if } p < p^* \text{ and } X \geq T, \\ \text{opt}(p) - \bar{u} & \text{if } p < p^* \text{ and } X < T, \\ 0 & \text{if } p \geq p^*; \end{cases} \\ \text{missing the target} &= \begin{cases} \text{opt}(p) - \bar{u} & \text{if } p \geq p^* \text{ and } X < T, \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

Below the pivot, a game that reaches the target loses the difference between keeping everything and paying the fair share, plus any overpayment. At or above the pivot, a game that reaches the target loses only its overpayment, and a game that misses loses everything it could have kept beyond the lottery. The shares reported in the paper are sums of each part over a model's games divided by its total loss.

S2.6 How much risk aversion would explain paying at low risk

Take a player with constant relative risk aversion γ , whose utility of final cash w is $w^{1-\gamma}/(1-\gamma)$ for $\gamma < 1$, and compare two plans: pay the fair share and reach the target, which leaves 20 for certain, or pay nothing and miss, which leaves 40 with probability $1 - p$ and 0 otherwise. The player prefers the certain 20 if and only if $20^{1-\gamma} \geq (1 - p) 40^{1-\gamma}$, that is

$$\gamma \geq 1 - \log_2 \frac{1}{1 - p}.$$

At $p = 0.1$ this needs γ of at least 0.85. For $\gamma \geq 1$ the utility of 0 is unbounded below, so any positive risk makes the certain 20 preferable; this is why the paper speaks of strong risk aversion. At $p = 0$ the two plans leave 20 and 40 for certain, and no increasing utility prefers 20.

Table S2: The four kinds of scripted partner.

Name	Rule for each scripted seat	Partners' pool over ten rounds
Always 0	Pays 0 in every round	0
Always 2	Pays 2 in every round (the fair share)	100
Always 4	Pays 4 in every round	200
Copy the group	Pays 2 in round 1; afterwards pays the allowed amount nearest to the average of the other five seats' contributions in the previous round, ties going to the lower amount	100, whatever the model does (see text)

S2.7 Scripted partners and their best responses

In the scripted design, seat 1 holds the model and seats 2 to 6 follow one of four fixed rules (Table S2). The model is not told that its partners are scripted.

Best responses. Beside partners who always pay 0, the pool is at most 40, so the target cannot be reached and the model earns $(1 - p)(40 - x_1)$: the best response is to pay nothing. Beside partners who always pay 4, the target is reached whatever the model does and it earns $40 - x_1$: again the best response is to pay nothing. Beside partners who always pay 2, the target is reached if and only if $x_1 \geq 20$, so the model compares 20 for certain with $(1 - p)40$ from paying nothing: the best response is 0 below p^* , 20 above it, and both at p^* . A partner who copies the group averages four scripted seats paying 2 and the model's last move $a \in \{0, 2, 4\}$, which gives $(8 + a)/5 \in \{1.6, 2, 2.4\}$, and the nearest allowed amount is always 2. These partners therefore never leave 2, and their best response is the same as for partners who always pay 2. The analysis code confirms all four best responses by trying every one of the 59,049 possible sequences of the model's own moves at each risk level (Table S3).

Table S3: Best response of the model's seat to each kind of partner, found by trying every sequence of its own moves. Top: the best total over ten rounds. Bottom: the expected payoff it earns. Beside partners who always pay 2 or copy the group, any sequence that adds up to 20 is a best response above p^* .

Partners	Risk p				
	0.1	0.3	0.5	0.7	0.9
<i>Best-response total over ten rounds</i>					
Always 0	0	0	0	0	0
Always 2	0	0	0 or 20	20	20
Always 4	0	0	0	0	0
Copy the group	0	0	0 or 20	20	20
<i>Best expected payoff</i>					
Always 0	36.0	28.0	20.0	12.0	4.0
Always 2	36.0	28.0	20.0	20.0	20.0
Always 4	40.0	40.0	40.0	40.0	40.0
Copy the group	36.0	28.0	20.0	20.0	20.0

When the outcome is settled. Paying is dominated once the history shows that more payment cannot change the outcome. Beside partners who always pay 4, the outcome is settled after the first round in which the pool reaches 120. The partners alone add 20 per round, so this happens after round 6 at the latest, and after round 5 if the model paid 4 in each of the first five rounds. Beside partners who always pay 0, the outcome is settled after the first round t in which the pool plus 24 per remaining round, the most six seats could still add, stays below 120. The pool then equals the model's own total, which is at most $4t$, so this happens after round 6 unless the model paid 4 in each of the first six rounds, and after round 7 in that case. Table S25 in Section S9 gives the observed settlement rounds.

S3 Models and protocol

S3.1 Models

Every game was played through one model-serving interface, with the same settings for all five models (Table S4).

Table S4: The five models. The paper uses the names in the first column.

Name in the paper	Model	Provider	Identifier
Claude Haiku 4.5	Claude Haiku 4.5	Anthropic	claude-haiku-4-5-20251001
Gemini 3.5 Flash-Lite	Gemini 3.5 Flash-Lite	Google	gemini-3.5-flash-lite
GPT-5.6 Luna	GPT-5.6 Luna	OpenAI	gpt-5.6-luna
Qwen3-235B	Qwen3-235B-A22B-Instruct-2507	Alibaba	qwen3-235b-a22b-instruct-2507
Grok 4.20	Grok 4.20, non-reasoning	xAI	grok-4.20-0309-non-reasoning

S3.2 How a round is played

In each round every seat that holds a model receives its own prompt in a separate call with no memory of earlier calls. The prompt contains the rules, the seat’s position, the current round, the seat’s remaining money and every seat’s contribution in every earlier round, listed by fixed position (Section S4). The six calls of a round see the same history, so decisions are simultaneous. The running pool is not printed. No seat is given a persona, and all prompts are in English.

Sampling uses temperature 0.7, or 0 in the temperature-zero condition, and an output limit of 3,000 tokens. Each call carries a sampling seed derived from the game, the seat and the round. The service accepts the seed but does not guarantee identical text, so replaying a game does not reproduce its transcripts (Section S5.4).

S3.3 How replies are read

The move is read from the last line of the reply that starts with **CONTRIBUTION:** and carries one of the allowed amounts. A reply without such a line falls back to the last allowed amount it names. Two situations trigger a new request, with a fresh sampling seed, up to three times: a reply that names no allowed amount at all, and a reply whose length reached the output limit. A reply that reached the limit is asked again with a four times larger limit, up to 8,000 tokens, because asking again with the same limit cuts the reply at the same place; such a reply is never counted as readable, since any number in it may come from unfinished reasoning. A reply that is still unusable after the retries is counted as unreadable. The data record, for every seat and game, how many replies were unreadable or lacked a **CONTRIBUTION:** line. No reply was unreadable. Across all 187,000 model decisions, 2 replies lacked the line, both from Grok 4.20 under the first rewording (Table S5), and their moves were read from the rest of the reply. Both games exceeded the target by more than a single move can change, so neither move affects an outcome, and every other move comes from a **CONTRIBUTION:** line.

In-game questions are read the same way from a final **ANSWER:** line (Section S8). An unreadable answer counts as wrong.

S4 Prompts, word for word

Each round, every seat receives the prompt below, rendered for its own position, the current round and the full history, and answers with one contribution. The texts are generated from the program that played the games, not retyped, for one example state.

S4.1 Base prompt

The prompt one seat receives in round 3 at $p = 0.9$ (the history is an example). The two equal-split phrases, which the no-hint prompt deletes, are in bold.

You are Player_3, player 3 of 6, in a group experiment that simulates a collective-risk social dilemma. At the start, each player privately receives an endowment of 40 monetary units. Your final cash payoff depends on the rules below. The game lasts exactly 10 rounds.

Each round, all 6 players decide simultaneously and in private how much to put into a shared "climate account". The only amounts allowed are 0, 2, 4.

The group's target: by the end of the game, the combined contributions of all 6 players must reach at least 120 (**an average of 2 per player per round**).

- If the group reaches 120: every player keeps, in cash, whatever they did not contribute (**for example, contributing 2 every round leaves you 20 at the end**).
- If the group does not reach 120: the computer runs a lottery: with probability 90% the group is hit by a disaster and every player loses all of their remaining money (final cash payoff 0); with the remaining 10% probability, each player keeps their remaining money.
- Money put into the climate account is gone for good and is never refunded, whether or not the target is reached.

After each round, all 6 contributions for that round are revealed to everyone. Names are hidden, but each player always appears in the same fixed position. Players cannot talk to or message each other.

Current state:

- Round 3 of 10.
- Your remaining money: 36.

Previous rounds (each player keeps the same position; you are position 3):

Round 1: P1=2, P2=2, P3(you)=2, P4=2, P5=4, P6=2.

Round 2: P1=2, P2=0, P3(you)=2, P4=2, P5=4, P6=2.

Now decide your contribution for round 3. Allowed choices: 0, 2, 4.

Output only your decision as a final line:

CONTRIBUTION: <one of 0, 2, 4>

S4.2 No hint

Identical to the base prompt except for the two lines below.

Base: The group's target: by the end of the game, the combined contributions of all 6 players must reach at least 120 (an average of 2 per player per round).

- If the group reaches 120: every player keeps, in cash, whatever they did not contribute (for example, contributing 2 every round leaves you 20 at the end).

Changed to: The group's target: by the end of the game, the combined contributions of all 6 players must reach at least 120.

- If the group reaches 120: every player keeps, in cash, whatever they did not contribute.

S4.3 Neutral wording

Identical to the base prompt except for the lines below. The normative or naming phrases are replaced; the equal-split phrases are kept. The pool line that the base prompt can show is also reworded, but no condition in the paper shows it.

Base: You are Player_3, player 3 of 6, in a group experiment that simulates a collective-risk social dilemma.

Changed to: You are Player_3, player 3 of 6, in a group experiment.

Base: Each round, all 6 players decide simultaneously and in private how much to put into a shared "climate account". The only amounts allowed are 0, 2, 4.

Changed to: Each round, all 6 players decide simultaneously and in private how much to put into a shared account. The only amounts allowed are 0, 2, 4.

Base: The group's target: by the end of the game, the combined contributions of all 6 players must reach at least 120 (an average of 2 per player per round).

Changed to: The threshold: the outcome depends on whether, by the end of the game, the combined contributions of all 6 players reach at least 120 (an average of 2 per player per round).

Base: - If the group does not reach 120: the computer runs a lottery: with probability 90% the group is hit by a disaster and every player loses all of their remaining money (final cash payoff 0); with the remaining 10% probability, each player keeps their remaining money.
- Money put into the climate account is gone for good and is never refunded, whether or not the target is reached.

Changed to: - If the group does not reach 120: the computer runs a lottery: with probability 90% every player loses all of their remaining money (final cash payoff 0); with the remaining 10% probability, each player keeps their remaining money.
- Money put into the shared account is gone for good and is never refunded, whether or not the threshold is reached.

S4.4 Neutral wording, goal stated

The neutral-wording prompt with one opening paragraph added. Full text:

You are a participant in a paid economics experiment. The only thing that matters to you is your own final cash payoff. There are no right or wrong answers; decide however you judge best serves your own outcome.

You are Player_3, player 3 of 6, in a group experiment.

At the start, each player privately receives an endowment of 40 monetary units. Your final cash payoff depends on the rules below. The game lasts exactly 10 rounds.

Each round, all 6 players decide simultaneously and in private how much to put into a shared account. The only amounts allowed are 0, 2, 4.

The threshold: the outcome depends on whether, by the end of the game, the combined contributions of all 6 players reach at least 120 (an average of 2 per player per round).

- If the group reaches 120: every player keeps, in cash, whatever they did not contribute (for example, contributing 2 every round leaves you 20 at the end).
- If the group does not reach 120: the computer runs a lottery: with probability 90% every player loses all of their remaining money (final cash payoff 0); with the remaining 10% probability, each player keeps their remaining money.
- Money put into the shared account is gone for good and is never refunded, whether or not the threshold is reached.

After each round, all 6 contributions for that round are revealed to everyone. Names are hidden, but each player always appears in the same fixed position. Players cannot talk to or message each other.

Current state:

- Round 3 of 10.
- Your remaining money: 36.

Previous rounds (each player keeps the same position; you are position 3):

Round 1: P1=2, P2=2, P3(you)=2, P4=2, P5=4, P6=2.

Round 2: P1=2, P2=0, P3(you)=2, P4=2, P5=4, P6=2.

Now decide your contribution for round 3. Allowed choices: 0, 2, 4.

Output only your decision as a final line:

CONTRIBUTION: <one of 0, 2, 4>

S4.5 Two rewordings

Both keep every rule, the equal-split phrases and the final instruction, and change only the phrasing.

Rewording 1.

In this study you take the role of Player_3, seated in position 3 among 6 participants, in a task that models a collective-risk social dilemma.

Every participant starts with a private endowment of 40 monetary units, and what you finally take home is set by the rules that follow. There are exactly 10 rounds, no more and no fewer.

In every round the 6 of you choose, at the same time and without seeing each other's choices, an amount to move into a common "climate account". The permitted amounts are 0, 2, 4.

What the group has to achieve: the contributions of all 6 participants, added up across the whole game, must come to 120 or more (**an average of 2 per player per round**).

- Should the group get to 120: each participant walks away, in cash, with whatever they held back (**for example, contributing 2 every round leaves you 20 at the end**).

- Should the group fall short of 120: a lottery is drawn. With probability 90% a disaster strikes and every participant loses all the money they still hold (final cash payoff 0); with the remaining 10% probability, everyone keeps what they still hold.

- Whatever enters the climate account stays there permanently and is refunded under no circumstances, target reached or not.

Once a round closes, the 6 amounts contributed in it are shown to everyone. Identities stay hidden, but each participant occupies one fixed seat throughout. Participants cannot talk to or message one another.

Where things stand:

- This is round 3 out of 10.
- Money still in your hands: 36.

Earlier rounds (seats never change; yours is position 3):

Round 1: P1=2, P2=2, P3(you)=2, P4=2, P5=4, P6=2.

Round 2: P1=2, P2=0, P3(you)=2, P4=2, P5=4, P6=2.

Now decide your contribution for round 3. Allowed choices: 0, 2, 4.

Output only your decision as a final line:

CONTRIBUTION: <one of 0, 2, 4>

Rewording 2.

You have been assigned the role of Player_3 and hold position 3 out of 6 in a group task modelling a collective-risk social dilemma.

Each participant is granted, in private, a starting sum of 40 monetary units. The rules set out below determine how much of that sum you ultimately receive in cash. Play continues for precisely 10 rounds.

Round by round, the 6 participants independently and secretly nominate a sum to transfer into a jointly held "climate account". The admissible sums are 0, 2, 4, and no others.

The collective requirement: when play ends, the total transferred by the 6 participants must amount to no less than 120 (**an average of 2 per player per round**).

- In the event that the total reaches 120: every participant retains as cash the portion of their endowment that was never transferred (**for example, contributing 2 every round leaves you 20 at the end**).

- In the event that the total falls below 120: the computer conducts a draw. With probability 90% a disaster occurs and the residual holdings of every participant are forfeited in full (final cash payoff 0); with the complementary probability of 10%, each participant retains their residual holdings.

- Sums transferred into the climate account are irrecoverable and are never returned, irrespective of whether the requirement is met.

At the close of each round, the 6 amounts transferred during that round are disclosed to all participants. Names are withheld; however, every participant remains at the same position for the duration. Communication of any kind between participants is not permitted.

Present position:

- Round 3 of 10 is underway.
- Funds you still retain: 36.

Record of previous rounds (positions are fixed; yours is position 3):

Round 1: P1=2, P2=2, P3(you)=2, P4=2, P5=4, P6=2.

Round 2: P1=2, P2=0, P3(you)=2, P4=2, P5=4, P6=2.

Now decide your contribution for round 3. Allowed choices: 0, 2, 4.

Output only your decision as a final line:

CONTRIBUTION: <one of 0, 2, 4>

S5 Designs, data and statistics

S5.1 Games in each design

Table S5 counts the games of every design. Every model plays the same number of games in every design, and every cell (one condition, one risk level, one model or pair of models) holds 10 games, numbered by a repetition index from 1 to 10. The scripted-partner and two-model designs use the base prompt.

Table S5: Games per design. *Games per model* counts the games in which a model holds at least one seat; in the design with two models at one table each model sits in four of the ten pairs. *Model decisions* counts one decision per model seat per round (the scripted seats are not counted). *Without the marker*: replies with no CONTRIBUTION: line, whose move was read from the rest of the reply (Section S3.3); no reply was unreadable.

Design	Risk levels p	Games per model	Total games	Model decisions	Without the marker
Self-play grid	0, 0.1, ..., 1	110	550	33,000	0
Rerun (in-game questions)	0.1, 0.5, 0.9	30	150	9,000	0
No hint	0.1, 0.5, 0.9	30	150	9,000	0
Rewording 1	0.1, 0.9	20	100	6,000	2
Rewording 2	0.1, 0.9	20	100	6,000	0
Temperature zero	0.1, 0.9	20	100	6,000	0
Neutral wording	0, 0.1, 0.9	30	150	9,000	0
Goal stated	0, 0.1, 0.9	30	150	9,000	0
One model, five scripted partners					
Partners always pay 0	0.1, 0.3, ..., 0.9	50	250	2,500	0
Partners always pay 2	0.1, 0.3, ..., 0.9	50	250	2,500	0
Partners always pay 4	0.1, 0.3, ..., 0.9	50	250	2,500	0
Partners copy the group	0.1, 0.3, ..., 0.9	50	250	2,500	0
Two models at one table	0.1, 0.5, 0.9	600	1,500	90,000	0
Total			3,950	187,000	2

In the design with two models at one table, the ten pairs are all unordered pairs of the five models. The first-named model of a pair holds one to five seats and the second model holds the rest. Each model's seats form one contiguous block. The block positions are the same in every game of a cell (one pair, one split, one risk level) and the order of the two blocks alternates across cells, so every model holds every seat equally often only when cells are pooled.

S5.2 The lottery and why the paper uses expected payoffs

In the game engine, the catastrophe lottery of a game is one uniform draw that depends only on the repetition index, and a missed game ends in catastrophe if the draw is below p . The whole data set therefore uses the same ten draws, in every cell, model and design (Table S6), and all ten fall below 0.47. As a result, every missed game at $p \geq 0.5$ ended in catastrophe, and missed games at low risk ended in catastrophe more often than the risk implies: in the self-play grid, 143 games missed the target and 96 of them ended in catastrophe, against 69 expected from their risk levels. Realised payoffs would therefore punish models that miss the target more than chance would. The agents never see the draw, so their play is not affected, and every payoff in the paper is the expected payoff (1) given the contributions.

Table S6: The ten lottery draws. A missed game ends in catastrophe when its risk p exceeds the draw; the third column gives the risk levels of the self-play grid at which that happens. The last three columns count missed games in the self-play grid by repetition, and the catastrophes realised and expected (the sum of p over those games).

Repetition	Draw	A missed game ends in catastrophe at	Self-play grid		
			Missed games	Catastrophes Realised	Expected
1	0.417	$p \geq 0.5$	16	9	8.0
2	0.463	$p \geq 0.5$	16	9	8.0
3	0.354	$p \geq 0.4$	14	8	6.6
4	0.049	$p \geq 0.1$	17	15	9.1
5	0.062	$p \geq 0.1$	12	10	5.5
6	0.192	$p \geq 0.2$	13	10	6.2
7	0.373	$p \geq 0.4$	13	7	5.5
8	0.355	$p \geq 0.4$	13	9	6.7
9	0.360	$p \geq 0.4$	14	7	5.9
10	0.134	$p \geq 0.2$	15	12	7.6
All			143	96	69.1

S5.3 Statistical methods

- **Unit.** The game. Seats of one game share a history and a lottery, so they are not independent. A model’s value in a game is the mean over the seats it holds.
- **Intervals.** 95% percentile bootstrap over games with 5,000 resamples (2,000 for the tables of Section S12). For self-play and the prompt controls, a quantity that pools several risk levels resamples games within each risk level, so each level keeps its weight. Intervals for the scripted and two-model designs, including Figure 4 of the paper, resample games pooled across risk levels without stratification; every such pool holds the same number of games at each risk level. For a ratio, such as the share of $\text{opt}(p)$ lost, numerator and denominator are resampled together.
- **Tests.** Two-sided permutation tests on the difference in means, with 5,000 permutations, reporting $(h + 1)/(n + 1)$ for h permuted differences at least as large as the observed one out of n . When two conditions are compared over several risk levels, condition labels are permuted within each risk level. The paper marks a prompt change in bold when the shift is at least 2 units and $P < 0.01$.
- **Exact sign-flip tests.** For Qwen3-235B against its tablemates in each cell of ten games, the one-sided P is the share of all 2^{10} sign patterns of the per-game payoff gaps whose mean is at most the observed mean.
- **Variance shares.** For the scripted design, the variance of a model’s total over the three fixed-amount partner kinds, five risk levels and ten games is split with balanced two-way sums of squares into partner kind, risk, their interaction and a residual.
- **Selection.** Uncertainty comes from 1,000 bootstrap replicates that resample the games of every mixed cell and every self-play cell before the whole selection analysis is recomputed (Section S11).
- **Seeds.** Every resampling uses a fixed random seed, so rerunning the analysis code reproduces every number in the paper and in this supplement exactly.

S5.4 Reproducibility

The prompts, configurations, seeds and code will be released with the final version. The design is reproducible: the lottery draws and the seat layouts depend only on the repetition index and the cell. Model text is not. The model service accepts a sampling seed but does not return identical text for it, and even temperature zero is not deterministic: a ten-game cell at temperature zero contained 5.5 distinct game records on average, against 6.6 at temperature 0.7, where two games count as the same only if all six seats made the same ten moves. Replaying a cell is therefore a new sample of the same design, not a copy of the old one.

S6 Self-play across risk

This section supports Section 4.1.1 of the paper, its Table 3 and its Figure 2. All six seats hold the same model, with the base prompt, at eleven risk levels from 0 to 1 and ten games per level, so each model plays 110 games and fills 660 seats.

S6.1 Numbers behind the main text

Statement in the paper, made precise	Value
Largest gap between the mean total above and below the pivot ($p \geq 0.5$ against $0 < p < 0.5$), over the four models other than Grok 4.20	0.2
the same gap for Grok 4.20	4.4
Smallest coefficient of relative risk aversion that prefers a certain fair share to paying nothing at $p = 0.1$ (Section S2.6)	0.85
Gemini 3.5 Flash-Lite seats that play the exact split, whole grid (%)	99.7
Models whose mean total at $p = 0$ is above half the fair share	four
Seats that pay anything at $p = 0$	240 of 300
GPT-5.6 Luna's mean total at $p = 0$	0.0
Grok 4.20: mean round in which the pool reaches 120; units per seat paid after that	6.7; 13.0
Qwen3-235B: decisions paying 2 in rounds 1 to 9 (%); seats paying 0 in round 10 (%)	99; 95
Qwen3-235B: games reaching the target, of 110	2
Qwen3-235B: games reaching it had every seat repeated its round-9 move (%)	100
Qwen3-235B: games whose pool before round 10 is short of 120 by more than the 4 units one seat can add, of 110; most common pool before round 10	106; 108
GPT-5.6 Luna: games missing the target at $p > 0$, of 100	23
All models: missed games at $p > 0$ that were at or ahead of the fair pace through round 9	121 of 121
Claude Haiku 4.5: seats paying 0 in round 10 when the pool already holds 120 (%; 49 games); when it does not (%; 61 games)	89; 12
Share of the attainable payoff the panel loses (%)	42 [41, 43]
Gemini 3.5 Flash-Lite; Qwen3-235B; Grok 4.20 (%)	21; 57; 80

S6.2 What a seat pays at each risk level

Table S7 gives the mean seat total in each cell of the grid, under the optimum. Outside $p = 0$, every cell of Claude Haiku 4.5, Gemini 3.5 Flash-Lite, GPT-5.6 Luna and Qwen3-235B lies close to the fair share of 20, whether the optimum is 0 or 20. Grok 4.20 pays far more at every level, and its highest values are at zero risk and at the highest risk levels, so its payment does not rise steadily with the risk.

Means can hide a mixture of very different seats, so Figure S1 shows the whole distribution of seat totals. Gemini 3.5 Flash-Lite puts almost every seat on exactly 20, at every risk level. GPT-5.6 Luna does the same except at zero risk, where every seat pays 0. Claude Haiku 4.5 spreads around 20, Qwen3-235B concentrates just below 20, which is its habit of skipping the last payment, and Grok 4.20 spreads from 20 to the full 40.

Table S7: Mean seat total (units out of 40) in self-play, ten games per cell. The top row is the optimum of Proposition 1.

Model	Catastrophe probability p										
	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
<i>Optimum</i>	0	0	0	0	0	0 or 20	20	20	20	20	20
Claude Haiku 4.5	20.9	20.6	21.0	21.0	21.2	21.1	21.6	20.6	21.4	21.2	21.2
Gemini 3.5 Flash-Lite	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0
GPT-5.6 Luna	0.0	20.0	20.1	20.0	19.9	19.9	20.1	19.7	20.0	19.8	19.8
Qwen3-235B	18.0	18.4	18.9	18.4	18.3	18.6	18.1	18.5	18.2	18.4	18.0
Grok 4.20	37.4	31.9	28.6	33.3	34.7	33.6	34.5	34.2	38.8	38.2	39.9

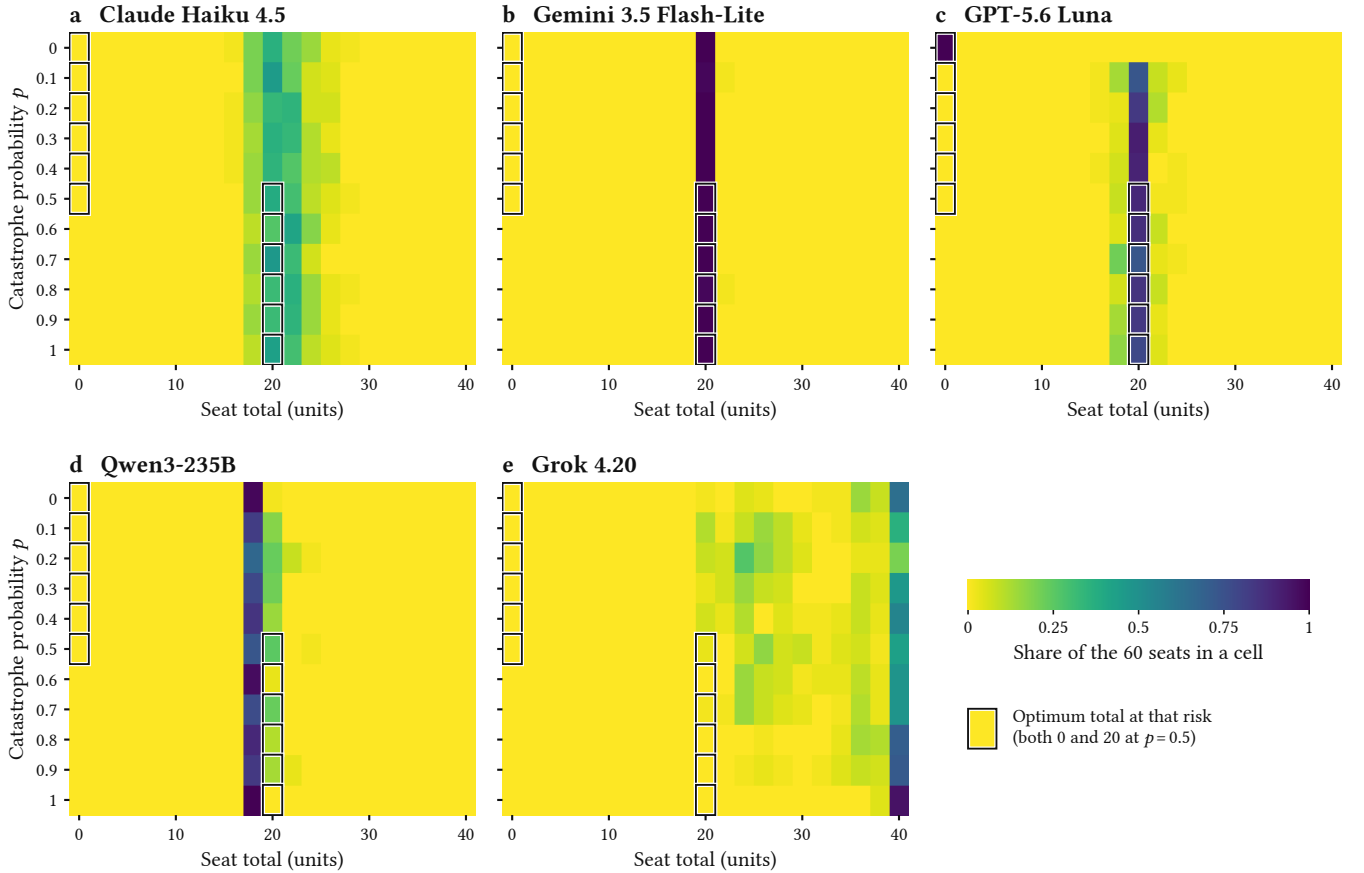


Figure S1: Distribution of seat totals in self-play. In each panel a row is one risk level and a column one possible total; the colour gives the share of that cell's 60 seats with that total. Outlined cells mark the optimal total at that risk level (both 0 and 20 at $p = 0.5$).

S6.3 Outcomes at each risk level

Table S8 gives three outcomes for every cell. Gemini 3.5 Flash-Lite and Grok 4.20 reach the target at every risk level, and so does Claude Haiku 4.5 at every positive level. Qwen3-235B almost never does. The share of the attainable payoff that is lost shows the cost of the default at low risk: the three fair-share models lose close to half of it at $p = 0.1$, where the optimum is to keep everything. Above the pivot, Claude Haiku 4.5 and Gemini 3.5 Flash-Lite lose little, while GPT-5.6 Luna still loses part of it through the games that miss the target.

Table S8: Self-play outcomes by risk level, ten games per cell. (a) Games reaching the target. (b) Seats paying 2 in all ten rounds. (c) Share of the attainable expected payoff $\text{opt}(p)$ lost; the Panel row pools the five models. *All* pools the grid and matches Table 3 of the paper.

(a) Target reached (% of games)												
Model	Catastrophe probability p											All
	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1	
Claude Haiku 4.5	80	100	100	100	100	100	100	100	100	100	100	98
Gemini 3.5 Flash-Lite	100	100	100	100	100	100	100	100	100	100	100	100
GPT-5.6 Luna	0	80	90	90	70	80	90	50	90	70	60	70
Qwen3-235B	0	0	20	0	0	0	0	0	0	0	0	2
Grok 4.20	100	100	100	100	100	100	100	100	100	100	100	100

(b) Exact split (% of seats)												
Model	Catastrophe probability p											All
	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1	
Claude Haiku 4.5	23	8	20	13	15	20	17	23	15	15	12	17
Gemini 3.5 Flash-Lite	100	98	100	100	100	100	100	100	98	100	100	99.7
GPT-5.6 Luna	0	68	73	92	90	87	83	67	80	82	75	72
Qwen3-235B	0	2	2	3	8	13	0	12	3	2	0	4
Grok 4.20	0	10	3	0	3	0	0	0	0	0	0	2

(c) Share of $\text{opt}(p)$ lost (%)												
Model	Catastrophe probability p											All
	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1	
Claude Haiku 4.5	52	46	41	32	22	6	8	3	7	6	6	26
Gemini 3.5 Flash-Lite	50	45	38	29	17	0	0	0	0.2	0	0	21
GPT-5.6 Luna	0	45	39	31	26	10	7	35	9	27	40	25
Qwen3-235B	45	46	45	46	46	47	56	68	78	89	100	57
Grok 4.20	94	78	64	76	78	68	73	71	94	91	99.5	80
<i>Panel</i>	48	52	45	43	38	26	29	35	38	43	49	42

S6.4 Where the payoff goes

Table S9 adds intervals to Table 3 of the paper and splits each model's loss into the three parts defined in Section S2.5. Claude Haiku 4.5 and Gemini 3.5 Flash-Lite lose almost only by paying below the pivot. GPT-5.6 Luna loses mostly there too, and the rest by missing the target above the pivot. Grok 4.20 loses mostly by paying past 120. Qwen3-235B loses in two nearly equal ways, paying below the pivot and missing above it.

Table S9: Self-play summary with intervals. At $p = 0$: mean seat total with its interval and the number of seats paying anything. *Payoff lost*: share of $\text{opt}(p)$ lost over the grid, with a bootstrap interval that resamples games within each risk level. *Share of the loss*: the three parts of Section S2.5 as shares of each model’s total loss.

Model	At $p=0$		Seats paying	Payoff lost		Share of the loss (%)		
	Mean total	95% CI		%	95% CI	Below p^*	Past 120	Missed at $p \geq p^*$
Claude Haiku 4.5	20.9	[20.2, 21.6]	60/60	26	[25, 26]	83	17	0
Gemini 3.5 Flash-Lite	20.0	[20.0, 20.0]	60/60	21	[21, 21]	99.9	0.1	0
GPT-5.6 Luna	0.0	[0.0, 0.0]	0/60	25	[21, 28]	63	1	36
Qwen3-235B	18.0	[18.0, 18.1]	60/60	57	[57, 58]	45	<0.1	55
Grok 4.20	37.4	[35.9, 38.6]	60/60	80	[77, 84]	27	73	0
<i>Panel</i>	19.3	[18.9, 19.6]	240/300	42	[41, 43]	50	30	19

S6.5 The end of the game

Table S10 describes the last rounds, which is where the models differ most. Qwen3-235B pays 2 in almost every decision of rounds 1 to 9 and 0 in almost every round-10 decision, so its tables almost always end just short of 120. Had every seat repeated its round-9 move, all of them would have reached the target. Claude Haiku 4.5 stops in round 10 mostly when the pool already holds 120, which shows that the pool can be worked out from the history in the prompt. Grok 4.20 reaches 120 around round 7 and keeps paying. All of GPT-5.6 Luna’s missed games at positive risk were at or ahead of the fair pace until the last round.

Table S10: End-of-game behaviour in self-play (110 games per model; “n/a” marks a case with no games). The pool after round 9 is what the seats could have added up before deciding in round 10. *At or ahead of fair pace*: the pool reached at least $12t$ after every round t up to 9. The last row replays each game with every seat’s round-10 move replaced by its round-9 move.

	Claude Haiku 4.5	Gemini 3.5 Flash-Lite	GPT-5.6 Luna	Qwen3 235B	Grok 4.20
Seats paying 0 in round 10 (%)	46	0	20	95	2
when the pool after round 9 is at least 120 (%)	89	n/a	n/a	100	2
games in this case	49	0	0	1	105
when it is 96 to 118 (%)	12	0	13	95	10
games in this case	61	110	100	109	5
Decisions paying 2 in rounds 1 to 9 (%)	88	>99.9	90	99	27
Games reaching 120 (of 110)	108	110	77	2	110
mean round the pool first reached 120	9.5	10.0	10.0	9.5	6.7
units per seat paid after that round	0.1	0.0	0.0	0.0	13.0
Games missing 120 at $p > 0$ (of 100)	0	0	23	98	0
of which at or ahead of fair pace through round 9	0	0	23	98	0
Games reaching 120 as played (%)	98	100	70	2	100
had every seat repeated its round-9 move in round 10 (%)	99	100	85	100	100

S7 Prompt and decoding controls

This section supports Section 4.1.2 of the paper and its Table 4. Section S4 gives every prompt. All conditions are self-play; the rerun repeats the base prompt, with in-game questions asked in separate calls that never enter the game.

S7.1 Numbers behind the main text

Statement in the paper, made precise	Value
No hint, Qwen3-235B: games reaching the target (%), rerun then no hint	0; 100
No hint, Qwen3-235B: expected payoff per seat, rerun then no hint	10.8; 5.2
No hint, GPT-5.6 Luna: seats playing the exact split (%), rerun then no hint	83; 56
No hint: Gemini 3.5 Flash-Lite seats playing the exact split; Grok 4.20 seats paying 4 in every round (%)	97; 99
Rewordings and temperature zero: largest change in mean total among the four models other than Grok 4.20	1.0
Grok 4.20: mean total on the base grid; under the second rewording ($p = 0.1$ and 0.9)	35.1; 20.6
Qwen3-235B seats paying 0 in round 10 across prompts, lowest to highest (%)	41 to 100
Grok 4.20: mean total at $p = 0.9$ minus at $p = 0.1$, base grid (P)	6.2 (0.0094)
at temperature zero (P)	6.1 (0.0116)
in the rerun; under the second rewording	1.9; 1.0
Neutral wording, $p = 0$: models whose mean total is above half the fair share; seats paying anything	four; 245 of 300
Goal stated, $p = 0$: seats paying anything	132 of 300
Goal stated, $p = 0$: mean total of Claude Haiku 4.5; Gemini 3.5 Flash-Lite; Qwen3-235B	0.3; 1.4; 4.8
the same models on the base grid	20.9; 20.0; 18.0
Goal stated, $p = 0.1$: mean total of Claude Haiku 4.5; Gemini 3.5 Flash-Lite	19.8; 20.2
Goal stated, GPT-5.6 Luna: mean total at $p = 0.1$; at $p = 0.9$	7.7; 20.1
Goal stated, Qwen3-235B: mean total at $p = 0.1$; at $p = 0.9$; games reaching the target	6.4; 10.1; 0 of 30
Distinct game records per ten-game cell, temperature zero against 0.7	5.5 against 6.6

S7.2 Every change against the base grid

Table S11 gives, for every model and condition, the mean seat total at $p = 0.1$ and 0.9 with its shift from the base grid, the interval of the shift and the permutation test. These are the tests behind the bold entries of Table 4 in the paper. The four models other than Grok 4.20 barely move under the rerun, the rewordings and temperature zero. The large shifts come from removing the hint (Qwen3-235B and Grok 4.20 pay more), the second rewording (Grok 4.20 pays about the fair share), neutral wording (Grok 4.20 pays more) and the stated goal (GPT-5.6 Luna and Qwen3-235B pay less).

Table S11: Mean seat total pooled over $p = 0.1$ and 0.9 (20 games per entry) and its shift from the base grid, with a bootstrap interval that resamples games within each risk level and a permutation P that permutes condition labels within each risk level. Bold: shift of at least 2 units with $P < 0.01$, as in Table 4 of the paper. A printed shift can differ by 0.1 from the difference of the printed means, because both are rounded separately.

Model	Condition	Mean seat total	Shift from base grid	95% CI	P
Claude Haiku 4.5	Base grid	20.9			
	Rerun	21.3	0.4	[−0.2, 0.9]	0.2517
	No hint	21.7	0.7	[0.2, 1.2]	0.0114
	Rewording 1	21.3	0.4	[−0.1, 0.9]	0.1618
	Rewording 2	20.9	0.0	[−0.5, 0.5]	1.0000
	Temperature zero	21.2	0.2	[−0.3, 0.8]	0.4403
	Neutral wording	21.0	0.1	[−0.4, 0.6]	0.7013
	Goal stated	20.0	−0.9	[−1.4, −0.4]	0.0018
Gemini 3.5 Flash-Lite	Base grid	20.0			
	Rerun	20.0	0.0	[−0.1, 0.0]	1.0000
	No hint	20.0	0.0	[0.0, 0.1]	1.0000
	Rewording 1	20.0	0.0	[0.0, 0.1]	1.0000
	Rewording 2	20.0	0.0	[−0.1, 0.0]	0.5273
	Temperature zero	20.0	0.0	[−0.1, 0.0]	0.5339
	Neutral wording	20.0	0.0	[−0.1, 0.1]	1.0000
	Goal stated	20.1	0.1	[−0.1, 0.3]	0.4647
GPT-5.6 Luna	Base grid	19.9			
	Rerun	19.9	0.0	[−0.3, 0.2]	0.9140
	No hint	19.9	0.0	[−0.3, 0.3]	1.0000
	Rewording 1	19.9	0.0	[−0.3, 0.2]	1.0000
	Rewording 2	19.8	−0.1	[−0.5, 0.2]	0.6235
	Temperature zero	20.0	0.1	[−0.1, 0.3]	0.3411
	Neutral wording	19.4	−0.5	[−0.8, −0.2]	0.0034
	Goal stated	13.9	−6.0	[−7.6, −4.3]	<0.001
Qwen3-235B	Base grid	18.4			
	Rerun	18.5	0.1	[−0.1, 0.3]	0.3553
	No hint	34.7	16.3	[15.6, 16.9]	<0.001
	Rewording 1	19.4	1.0	[0.8, 1.2]	<0.001
	Rewording 2	18.0	−0.4	[−0.5, −0.3]	<0.001
	Temperature zero	18.3	−0.1	[−0.3, 0.0]	0.2282
	Neutral wording	18.7	0.4	[0.1, 0.6]	0.0086
	Goal stated	8.2	−10.2	[−10.7, −9.7]	<0.001
Grok 4.20	Base grid	35.1			
	Rerun	35.6	0.6	[−2.5, 3.5]	0.7077
	No hint	40.0	4.9	[2.9, 7.0]	<0.001
	Rewording 1	31.8	−3.2	[−6.4, −0.3]	0.0506
	Rewording 2	20.6	−14.5	[−16.5, −12.3]	<0.001
	Temperature zero	35.1	0.1	[−2.8, 2.9]	0.9814
	Neutral wording	39.7	4.7	[2.6, 6.8]	<0.001
	Goal stated	37.7	2.7	[0.6, 4.9]	0.0260

S7.3 The goal test at each risk level

Table S12 separates the three risk levels of the goal test. At $p = 0$, neutral wording leaves most payment in place, while the stated goal removes almost all of it for Claude Haiku 4.5 and Gemini 3.5 Flash-Lite and most of it for Qwen3-235B; Grok 4.20 keeps paying. At $p = 0.1$, the stated goal leaves Claude Haiku 4.5 and Gemini 3.5 Flash-Lite at the fair share. GPT-5.6 Luna and Qwen3-235B drop far below it, although most of their seats still pay something, and only GPT-5.6 Luna returns to the fair share at $p = 0.9$. Qwen3-235B never reaches the target under the goal. The seat counts show how widespread a change is: a low mean can come from a few large payers or from many small ones.

Table S12: Base grid, neutral wording and goal stated at $p = 0, 0.1$ and 0.9 (ten games per entry): mean seat total, games reaching the target, and seats paying anything (of 60). P : permutation test on the mean total, neutral wording against goal stated.

Model	Base grid			Neutral wording			Goal stated			<i>P</i> , neutral vs goal
	Mean total	Target (of 10)	Seats paying	Mean total	Target (of 10)	Seats paying	Mean total	Target (of 10)	Seats paying	
<i>p</i> = 0										
Claude Haiku 4.5	20.9	8	60	18.0	5	55	0.3	0	7	<0.001
Gemini 3.5 Flash-Lite	20.0	10	60	19.9	7	60	1.4	0	18	<0.001
GPT-5.6 Luna	0.0	0	0	0.4	0	10	0.2	0	5	0.2529
Qwen3-235B	18.0	0	60	18.1	0	60	4.8	0	42	<0.001
Grok 4.20	37.4	10	60	39.8	10	60	36.7	10	60	<0.001
<i>p</i> = 0.1										
Claude Haiku 4.5	20.6	10	60	20.9	10	60	19.8	7	60	0.0178
Gemini 3.5 Flash-Lite	20.0	10	60	20.0	10	60	20.2	9	60	0.4687
GPT-5.6 Luna	20.0	8	60	19.4	2	60	7.7	0	53	<0.001
Qwen3-235B	18.4	0	60	18.9	1	60	6.4	0	60	<0.001
Grok 4.20	31.9	10	60	39.4	10	60	36.7	10	60	<0.001
<i>p</i> = 0.9										
Claude Haiku 4.5	21.2	10	60	21.2	10	60	20.2	9	60	0.0054
Gemini 3.5 Flash-Lite	20.0	10	60	20.0	10	60	20.0	10	60	1.0000
GPT-5.6 Luna	19.8	7	60	19.4	4	60	20.1	9	60	0.0280
Qwen3-235B	18.4	0	60	18.6	0	60	10.1	0	60	<0.001
Grok 4.20	38.2	10	60	40.0	10	60	38.7	10	60	<0.001

S7.4 How each model's signature changes

Each model has a signature on the base grid: Qwen3-235B skips the last payment, Gemini 3.5 Flash-Lite and GPT-5.6 Luna play the exact split, and Grok 4.20 often pays 4 in every round. Table S13 follows these signatures across conditions. The averages in Table S11 are stable, but the signatures are not. Qwen3-235B's last-round drop ranges from less than half of its seats to all of them. GPT-5.6 Luna plays the exact split far less often under neutral wording and the stated goal. Grok 4.20's rise with risk appears on the base grid and at temperature zero but not in the rerun of the same prompt, which is why the paper does not read it as a response to risk.

Table S13: Model signatures by condition, at $p = 0.1$ and 0.9 pooled (20 games per model). *Exact split*: seats paying 2 in all ten rounds. *Total at 0.9 minus at 0.1*: Grok 4.20’s mean seat total, with a permutation P . *Distinct games*: game records per ten-game cell that differ in at least one move, averaged over cells.

Condition	Qwen3 235B	Gemini 3.5 Flash-Lite	GPT-5.6 Luna	Grok 4.20			
	Pays 0 in round 10 (%)	Exact split (%)	Exact split (%)	Pays 4 in every round (%)	Total at 0.9 minus at 0.1	P	Distinct games
Base grid	97	99	75	54	6.2	0.0094	6.6
Rerun	92	100	84	60	1.9	0.4353	6.1
No hint	41	98	58	99	0.1	1.0000	6.1
Rewording 1	57	97	71	26	3.9	0.1230	7.9
Rewording 2	100	99	63	0	1.0	0.2158	4.6
Temperature zero	93	99	92	57	6.1	0.0116	5.5
Neutral wording	93	99	33	93	0.6	0.0126	6.4
Goal stated	100	73	32	49	2.0	0.0032	8.2

S7.5 Removing the hint

The no-hint prompt was played at $p = 0.1, 0.5$ and 0.9 , the same levels as the rerun, so Table S14 compares the two directly. Without the hint, no model pays significantly less on average, and two pay more. The hint mainly bought precision: fewer seats play the exact split without it, and Grok 4.20 moves almost entirely to paying 4 in every round. Qwen3-235B now pays enough to reach the target, but earns less, because it overpays.

Table S14: No hint against the rerun of the base prompt at $p = 0.1, 0.5$ and 0.9 (30 games per model in each). P : permutation test permuting condition labels within each risk level; panel rows also permute within each model.

Measure and model	Rerun	No hint	No hint minus rerun	P
Target reached (% of games)				
Claude Haiku 4.5	100	100	0	1.0000
Gemini 3.5 Flash-Lite	100	100	0	1.0000
GPT-5.6 Luna	80	57	-23	0.0880
Qwen3-235B	0	100	100	<0.001
Grok 4.20	100	100	0	1.0000
<i>Panel</i>	76	91	15	<0.001
Expected payoff per seat				
Claude Haiku 4.5	18.7	18.4	-0.3	0.2655
Gemini 3.5 Flash-Lite	20.0	19.9	-0.1	0.1128
GPT-5.6 Luna	18.3	16.8	-1.5	0.2096
Qwen3-235B	10.8	5.2	-5.7	<0.001
Grok 4.20	4.7	0.0	-4.6	<0.001
<i>Panel</i>	14.5	12.1	-2.4	<0.001
Mean seat total				
Claude Haiku 4.5	21.3	21.6	0.3	0.2589
Gemini 3.5 Flash-Lite	20.0	20.1	0.1	0.1202
GPT-5.6 Luna	19.9	19.8	-0.1	0.6269
Qwen3-235B	18.4	34.8	16.4	<0.001
Grok 4.20	35.3	40.0	4.6	<0.001
<i>Panel</i>	23.0	27.2	4.3	<0.001
Exact split (% of seats)				
Claude Haiku 4.5	15	1	-14	<0.001
Gemini 3.5 Flash-Lite	100	97	-3	0.1210
GPT-5.6 Luna	83	56	-27	0.0054
Qwen3-235B	6	0	-6	<0.001
Grok 4.20	0	0	0	1.0000
<i>Panel</i>	41	31	-10	<0.001
Pays 4 in every round (% of seats)				
Claude Haiku 4.5	0	0	0	1.0000
Gemini 3.5 Flash-Lite	0	0	0	1.0000
GPT-5.6 Luna	0	0	0	1.0000
Qwen3-235B	0	16	16	<0.001
Grok 4.20	57	99	42	<0.001
<i>Panel</i>	11	23	12	<0.001

S8 In-game questions

This section supports the value questions of Section 3.3 and the answers-against-play paragraph of Section 4.1.2 of the paper, and gives the full comparison of answers and play (Section S8.6). In the rerun games, seat 1 was asked ten questions in rounds 1, 5 and 10, about the state it had just decided on. Each question is a separate call that sees the decision prompt of that round with its final instruction replaced by the question, and its answer never enters the game. The answer is read from a final ANSWER: line; an unreadable answer would count as wrong. Each model answers 90 questions of each kind: 30 games, three rounds each.

S8.1 Numbers behind the main text

Statement in the paper, made precise	Value
Games with questions; questions asked; rule questions	150; 4,500; 3,600
Rule questions answered correctly (%)	100
Value comparison answered correctly (%): Claude Haiku 4.5; Gemini 3.5 Flash-Lite; GPT-5.6 Luna	100; 99; 84
Qwen3-235B; Grok 4.20	33; 46
These three models at $p = 0.1$, rounds 1 and 5: answers saying that paying nothing pays more (%)	97
The same seats' mean total at $p = 0.1$; at $p = 0.9$	20.1; 20.3
change, with interval	0.3 [-0.5, 1.0]
Qwen3-235B: answers saying that paying nothing pays more, all risk levels (%)	100
Grok 4.20: the same answer in round 1, all risk levels (%)	100
GPT-5.6 Luna: games whose value answer changes across rounds, of 30	13

S8.2 A question as the agent receives it

The prompt seat 1 receives in round 5 at $p = 0.9$ for the question that compares two strategies (the history is an example). It is the decision prompt of that round with the final instruction replaced; the replacement is in bold.

You are Player_1, player 1 of 6, in a group experiment that simulates a collective-risk social dilemma. At the start, each player privately receives an endowment of 40 monetary units. Your final cash payoff depends on the rules below. The game lasts exactly 10 rounds.

Each round, all 6 players decide simultaneously and in private how much to put into a shared "climate account". The only amounts allowed are 0, 2, 4.

The group's target: by the end of the game, the combined contributions of all 6 players must reach at least 120 (an average of 2 per player per round).

- If the group reaches 120: every player keeps, in cash, whatever they did not contribute (for example, contributing 2 every round leaves you 20 at the end).
- If the group does not reach 120: the computer runs a lottery: with probability 90% the group is hit by a disaster and every player loses all of their remaining money (final cash payoff 0); with the remaining 10% probability, each player keeps their remaining money.
- Money put into the climate account is gone for good and is never refunded, whether or not the target is reached.

After each round, all 6 contributions for that round are revealed to everyone. Names are hidden, but each player always appears in the same fixed position. Players cannot talk to or message each other.

Current state:

- Round 5 of 10.
- Your remaining money: 30.

Previous rounds (each player keeps the same position; you are position 1):

Round 1: P1(you)=2, P2=2, P3=2, P4=2, P5=4, P6=2.

Round 2: P1(you)=2, P2=0, P3=2, P4=2, P5=4, P6=2.

Round 3: P1(you)=2, P2=2, P3=2, P4=2, P5=2, P6=2.

Round 4: P1(you)=4, P2=2, P3=2, P4=0, P5=2, P6=2.

Now answer the following question about the game described above. This is a comprehension check, not a move: do not make any contribution decision.

Compare two strategies for the whole game. (A) You contribute 2 in every one of the 10 rounds and the group reaches the target, so you finish with 20 for certain. (B) You contribute 0 in every round and the group does not reach the target, so you keep 40 unless the disaster strikes. Which gives the higher AVERAGE final cash? Answer 1 if A is higher, 2 if B is higher, 0 if they are exactly equal.
 Answer with a single final line in exactly this format, and write nothing after it:
 ANSWER: <your answer>

S8.3 The ten questions

Each question is asked in rounds 1, 5 and 10 of every game, in a separate call. No question text depends on the round or the risk level.

Label	Question as asked	Answer format	Correct answer		
			$p = 0.1$	$p = 0.5$	$p = 0.9$
Rule questions					
Allowed amounts	Which contribution amounts are you allowed to choose each round?	A list of numbers	0, 2, 4	0, 2, 4	0, 2, 4
Starting endowment	How many monetary units did each player receive as their starting endowment?	A number	40	40	40
Group target	What combined total must all players reach by the end of the game to avoid the risk?	A number	120	120	120
Number of rounds	How many rounds does the game last in total?	A number	10	10	10
Disaster probability	If the group does NOT reach the target, what is the percent chance (0-100) that everyone loses all of their remaining money?	A number	10	50	90
Payoff after disaster	If the group fails to reach the target and the disaster does strike, what is your final cash payoff?	A number	0	0	0
Largest payment	What is the largest amount any single player may put into the climate account in one round?	A number	4	4	4
Smallest payment	What is the smallest amount any single player may put into the climate account in one round?	A number	0	0	0
Value questions					
Expected cash from paying nothing	Suppose you contribute 0 in every round and the group ends up NOT reaching the target. On average, over many such games, how much cash would you end up with? Give a single whole number.	A number	36	20	4
Two strategies compared	Compare two strategies for the whole game. (A) You contribute 2 in every one of the 10 rounds and the group reaches the target, so you finish with 20 for certain. (B) You contribute 0 in every round and the group does not reach the target, so you keep 40 unless the disaster strikes. Which gives the higher AVERAGE final cash? Answer 1 if A is higher, 2 if B is higher, 0 if they are exactly equal.	A code: 1, 2 or 0	2 (B)	0 (equal)	1 (A)

S8.4 Rule questions

Every rule question was answered correctly by every model in every round (Table S15). These questions ask for numbers printed in the prompt, so they show that the agents read the rules, not that they can use them.

Table S15: Rule questions: share answered correctly (%), with the number of answers in brackets.

Question	Claude Haiku 4.5	Gemini 3.5 Flash-Lite	GPT-5.6 Luna	Qwen3 235B	Grok 4.20	All models
Allowed amounts	100 (90)	100 (90)	100 (90)	100 (90)	100 (90)	100 (450)
Starting endowment	100 (90)	100 (90)	100 (90)	100 (90)	100 (90)	100 (450)
Group target	100 (90)	100 (90)	100 (90)	100 (90)	100 (90)	100 (450)
Number of rounds	100 (90)	100 (90)	100 (90)	100 (90)	100 (90)	100 (450)
Disaster probability	100 (90)	100 (90)	100 (90)	100 (90)	100 (90)	100 (450)
Payoff after disaster	100 (90)	100 (90)	100 (90)	100 (90)	100 (90)	100 (450)
Largest payment	100 (90)	100 (90)	100 (90)	100 (90)	100 (90)	100 (450)
Smallest payment	100 (90)	100 (90)	100 (90)	100 (90)	100 (90)	100 (450)
All rule questions	100 (720)	100 (720)	100 (720)	100 (720)	100 (720)	100 (3,600)

S8.5 The value comparison

Rule questions only ask for numbers shown in the prompt, so the paper relies on one value question, which compares two strategies for the whole game: (A) contribute 2 in every round and reach the target, finishing with 20 for certain; (B) contribute nothing and miss the target, keeping 40 unless the disaster strikes. The correct answer is B below p^* , A above it, and equal at p^* . A second value question could be read in two ways, so it is reported only here (Section S8.7).

Table S16 counts every answer to the value comparison. Claude Haiku 4.5 is always correct and Gemini 3.5 Flash-Lite is almost always correct; both always answer “equal” at $p = 0.5$. GPT-5.6 Luna is correct most of the time. Qwen3-235B answers that paying nothing is better in all 90 answers, so it is correct only at $p = 0.1$. Grok 4.20 leans the same way.

Table S16: The value comparison of Section 3.3 of the paper: number of answers of each kind, of 30 per model and risk level, and share correct. A: pay 2 in every round and reach the target; B: pay nothing and miss it.

Model	Risk	Correct answer	Answers saying			Correct (%)
			A higher	B higher	Equal	
Claude Haiku 4.5	$p = 0.1$	B	0	30	0	100
	$p = 0.5$	Equal	0	0	30	100
	$p = 0.9$	A	30	0	0	100
	All		30	30	30	100
Gemini 3.5 Flash-Lite	$p = 0.1$	B	0	30	0	100
	$p = 0.5$	Equal	0	0	30	100
	$p = 0.9$	A	29	1	0	97
	All		29	31	30	99
GPT-5.6 Luna	$p = 0.1$	B	2	28	0	93
	$p = 0.5$	Equal	0	6	24	80
	$p = 0.9$	A	24	2	4	80
	All		26	36	28	84
Qwen3-235B	$p = 0.1$	B	0	30	0	100
	$p = 0.5$	Equal	0	30	0	0
	$p = 0.9$	A	0	30	0	0
	All		0	90	0	33
Grok 4.20	$p = 0.1$	B	0	23	7	77
	$p = 0.5$	Equal	3	15	12	40
	$p = 0.9$	A	6	20	4	20
	All		9	58	23	46

Figure S2 and Table S17 split the same answers by round. Claude Haiku 4.5, Gemini 3.5 Flash-Lite and Qwen3-235B give the same answer in all three rounds of almost every game. GPT-5.6 Luna drifts in round 10 at $p = 0.5$. Grok 4.20 changes the most: it says that paying nothing is better in round 1 at every risk level, and moves toward the correct answer later in the game.

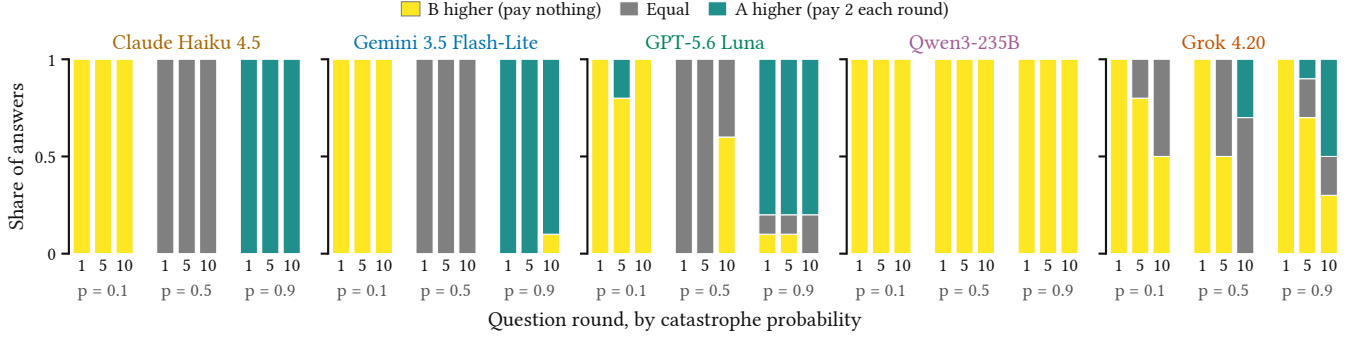


Figure S2: Answers to the value comparison by model, risk level and round (ten answers per bar).

Table S17: Value comparison by round: share of answers saying that B (paying nothing) is better, ten answers per entry; and the number of games whose answer is not the same in rounds 1, 5 and 10.

Model	Answers saying B (paying nothing) is higher (%)									Games whose answer changes			
	$p = 0.1$			$p = 0.5$			$p = 0.9$			By p (of 10)			All (of 30)
	R1	R5	R10	R1	R5	R10	R1	R5	R10	0.1	0.5	0.9	
Claude Haiku 4.5	100	100	100	0	0	0	0	0	0	0	0	0	0
Gemini 3.5 Flash-Lite	100	100	100	0	0	0	0	0	10	0	0	1	1
GPT-5.6 Luna	100	80	100	0	0	60	10	10	0	2	6	5	13
Qwen3-235B	100	100	100	100	100	100	100	100	100	0	0	0	0
Grok 4.20	100	80	50	100	50	0	100	70	30	6	10	8	24

S8.6 Answers against play

Every rule question is answered correctly. Claude Haiku 4.5 and Gemini 3.5 Flash-Lite answer the value question correctly almost every time, and GPT-5.6 Luna most of the time. All three change their answer with the risk (Figure S3a): at low risk they almost always say that keeping contributes more, and at high risk they mostly say the opposite. Their contributions barely change (Figure S3b). Qwen3-235B always says that keeping contributes more, and Grok 4.20 gives that answer in the first round, regardless of the risk. The question supplies two hypothetical outcomes rather than the partners' responses to a deviation, and it makes the comparison explicit, unlike the move request. Correct answers therefore do not establish knowledge of a best response, and the result is a gap between answers and play, not evidence about what an agent computes. A strongly risk-averse agent could give these answers and still contribute; contribution at $p = 0$ and after settlement provides the sharper test.

Table S18 gives the numbers behind Figure S3: for each model and risk level, the share of value answers saying that paying nothing is better and the total that the questioned seat pays in the same games, each with an interval. The figure plots the $p = 0.1$ and $p = 0.9$ rows with these intervals and joins them by an arrow; the *Shift* rows give that change, with intervals that resample the answer and the payment from the same game. For the three models that answer well, the answer moves by close to its full range while the payment does not move.

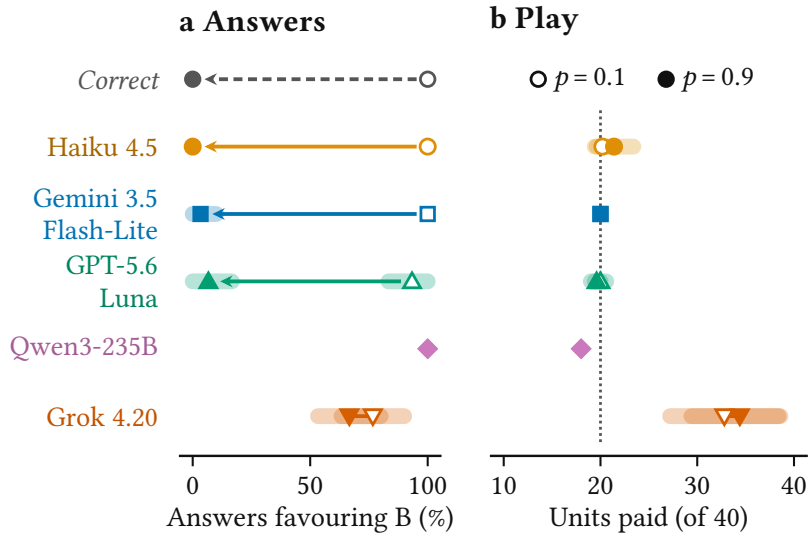


Figure S3: Answers and play at risk 0.1 (open) and 0.9 (filled), ten games per model and risk; arrows run from low to high risk, and shaded bars are 95% game-bootstrap intervals. (a) Share of answers favouring B; dashed: a fully correct switch. (b) Units contributed by the questioned seat out of 40; dotted: fair share. This compares answers and actions, not best responses; Table S18 gives the shifts with their intervals and all three risks.

Table S18: Answers and play of the questioned seat (ten games per risk level). *Shift*: drop in the share saying B from $p = 0.1$ to $p = 0.9$, and change in the units paid, with intervals that resample the games at the two risk levels independently.

Model	Risk	Share saying B (paying nothing) is higher		Units seat 1 pays	
		Mean	95% CI	Mean	95% CI
Claude Haiku 4.5	$p = 0.1$	1.00	[1.00, 1.00]	20.2	[19.4, 21.0]
	$p = 0.5$	0.00	[0.00, 0.00]	21.8	[20.8, 23.0]
	$p = 0.9$	0.00	[0.00, 0.00]	21.4	[19.6, 23.4]
	Shift	1.00	[1.00, 1.00]	1.2	[-0.6, 3.4]
Gemini 3.5 Flash-Lite	$p = 0.1$	1.00	[1.00, 1.00]	20.0	[20.0, 20.0]
	$p = 0.5$	0.00	[0.00, 0.00]	20.0	[20.0, 20.0]
	$p = 0.9$	0.03	[0.00, 0.10]	20.0	[20.0, 20.0]
	Shift	0.97	[0.90, 1.00]	0.0	[0.0, 0.0]
GPT-5.6 Luna	$p = 0.1$	0.93	[0.83, 1.00]	20.0	[19.4, 20.6]
	$p = 0.5$	0.20	[0.10, 0.30]	19.8	[19.4, 20.0]
	$p = 0.9$	0.07	[0.00, 0.17]	19.6	[19.0, 20.0]
	Shift	0.87	[0.73, 0.97]	-0.4	[-1.2, 0.4]
Qwen3-235B	$p = 0.1$	1.00	[1.00, 1.00]	18.0	[18.0, 18.0]
	$p = 0.5$	1.00	[1.00, 1.00]	18.0	[18.0, 18.0]
	$p = 0.9$	1.00	[1.00, 1.00]	18.0	[18.0, 18.0]
	Shift	0.00	[0.00, 0.00]	0.0	[0.0, 0.0]
Grok 4.20	$p = 0.1$	0.77	[0.63, 0.90]	32.8	[27.2, 38.4]
	$p = 0.5$	0.50	[0.40, 0.60]	29.6	[24.2, 35.2]
	$p = 0.9$	0.67	[0.53, 0.80]	34.4	[29.4, 38.6]
	Shift	0.10	[-0.10, 0.27]	1.6	[-5.8, 8.8]

S8.7 The second value question

The second value question asks how much cash a player ends with, on average, if it pays nothing and the group misses the target. The intended answer is $(1 - p) 40$ for the whole game. The question can also be read as starting from the moment it is asked, with the money the seat still holds, which gives $(1 - p)$ times that money. Table S19 shows that this second reading is the most common error: Claude Haiku 4.5 gives it in most of its round-5 answers, when it has already paid part of its endowment. Answers of 40, the cash kept if no disaster strikes, come only from Qwen3-235B at $p = 0.1$, where they are almost all of its answers. Answers of 0, the cash left if the disaster strikes, come mostly from GPT-5.6 Luna at $p = 0.5$ and 0.9 and from Grok 4.20 at $p = 0.9$, and Grok 4.20 gives other wrong answers to about a quarter of the questions at $p = 0.1$. Because a wrong answer can come from a reasonable reading rather than from a wrong computation, the paper does not use this question.

Table S19: The second value question, by model and risk level (30 answers per entry). *Correct (%)*: overall and in rounds 1, 5 and 10. *Wrong answers*, as shares of all answers: 40; 0; *Cash left*, which is $(1 - p)$ times the seat’s money at the time of asking, rounded; and any other number.

Model	Risk	Correct answer	Correct (%)				Wrong answers (% of all answers)			
			All	R1	R5	R10	40	0	Cash left	Other
Claude Haiku 4.5	$p = 0.1$	36	77	100	30	100	0	0	23	0
	$p = 0.5$	20	73	100	20	100	0	0	27	0
	$p = 0.9$	4	67	100	0	100	0	0	33	0
Gemini 3.5 Flash-Lite	$p = 0.1$	36	100	100	100	100	0	0	0	0
	$p = 0.5$	20	100	100	100	100	0	0	0	0
	$p = 0.9$	4	100	100	100	100	0	0	0	0
GPT-5.6 Luna	$p = 0.1$	36	93	100	80	100	0	0	7	0
	$p = 0.5$	20	77	50	80	100	0	17	7	0
	$p = 0.9$	4	80	70	70	100	0	20	0	0
Qwen3-235B	$p = 0.1$	36	3	0	10	0	93	0	0	3
	$p = 0.5$	20	100	100	100	100	0	0	0	0
	$p = 0.9$	4	97	100	100	90	0	3	0	0
Grok 4.20	$p = 0.1$	36	77	100	30	100	0	0	0	23
	$p = 0.5$	20	97	100	90	100	0	0	3	0
	$p = 0.9$	4	90	100	70	100	0	10	0	0

S9 One model beside scripted partners

This section supports Section 4.2.1 of the paper, including its Table 5; Figure S4 shows the same games round by round. Seat 1 holds the model and the five other seats follow one of the four rules of Table S2, at five risk levels with ten games each, so every model plays 50 games beside each kind of partner. Best responses are derived in Section S2.7.

S9.1 Numbers behind the main text

Statement in the paper, made precise	Value
Games beside partners who always pay 0 or always pay 4	500
of which the model pays nothing, which is the best response	0
smallest total the model pays in those games	2
Scripted games of any kind in which the model pays nothing	0 of 1,000
Scripted games that are exactly optimal (expected payoff equal to the best response)	186 of 1,000
total paid in every one of them	20
Share of the variation in the total explained by the partner kind, for Claude Haiku 4.5, Gemini 3.5 Flash-Lite, GPT-5.6 Luna and Qwen3-235B (%)	69 to 92
explained by risk and its interaction with the partner kind, largest of the four (%)	5
explained by risk and interaction for an exact best responder (%)	50
Largest change in a model's mean total between low and high risk beside fair-share partners	2.0
Grok 4.20: share explained by the partner kind; by risk and interaction (%)	3; 8
Grok 4.20: games opening with a payment of 4, of 200	103
Grok 4.20: share of the variation explained by its opening move (%)	74
Units paid after the outcome is settled, per game: GPT-5.6 Luna; Qwen3-235B; Grok 4.20	0.0; 10.1; 13.7
Qwen3-235B's mean total beside partners who always pay 0; always pay 2	29.9; 18.1
Qwen3-235B beside partners who always pay 2: games missing the target, of 50; shortfall	47; 2
Gemini 3.5 Flash-Lite in the same seat: games reaching the target, of 50	50

S9.2 What the models pay

Figure S4 shows, round by round, what each model pays beside partners who always pay 0 or always pay 4, the two kinds for which paying nothing is the best response. Look for the round at which each row turns light: Gemini 3.5 Flash-Lite and Claude Haiku 4.5 stop once the outcome is settled, GPT-5.6 Luna stops earlier, and Qwen3-235B and Grok 4.20 keep paying.

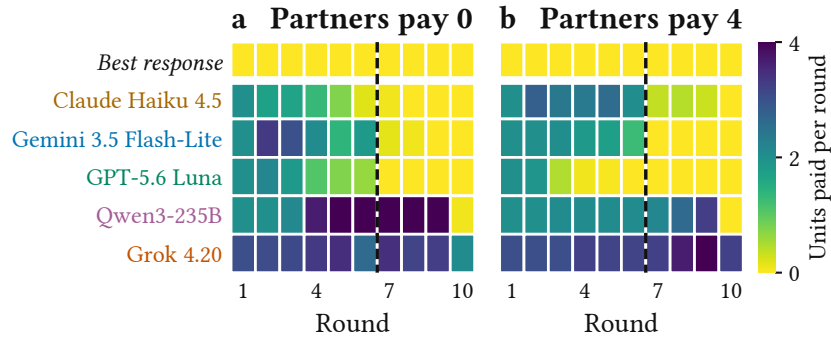


Figure S4: One model beside five scripted partners, pooled over five risk levels (50 games per row). A cell is the mean payment in one round; the top row is the best response, 0 in every round. The dashed rule follows round 6, after which the outcome is settled in most games.

Table S20 gives the mean total of the model's seat for every kind of partner and risk level, with the best response below each block. The last column is what a best responder would change: 20 units between low and high risk beside fair-share partners, and nothing beside the other two. No model comes close. Beside partners who always pay 0 or always pay 4, every model pays more than the best response of 0 in every game, and Qwen3-235B and Grok 4.20 pay the most, even though these partners make every unit a loss.

Figure S5 shows the same games round by round beside the two kinds of fair-share partner, where the best response switches at p^* . Gemini 3.5 Flash-Lite and GPT-5.6 Luna pay almost exactly 2 in every round at every risk level. Qwen3-235B pays 2 for nine rounds and 0 in the last, at every risk level, and Grok 4.20 pays more than 2 in every round. None of the rows changes between the low-risk and high-risk panels.

Table S20: Mean total paid by the model’s seat (ten games per entry). *All risks*: the mean over the five risk levels. Last column: the mean at $p = 0.7$ and 0.9 minus the mean at $p = 0.1$ and 0.3 . The best-response row gives the best total; “0 or 20” marks the tie at p^* .

Model	Risk p					All $p > 0.5$ minus	
	0.1	0.3	0.5	0.7	0.9	risks	$p < 0.5$
<i>Always 0</i>							
Claude Haiku 4.5	6.8	7.2	8.2	8.6	8.0	7.8	1.3
Gemini 3.5 Flash-Lite	12.6	11.8	14.2	14.2	16.6	13.9	3.2
GPT-5.6 Luna	6.8	6.0	9.0	8.6	12.0	8.5	3.9
Qwen3-235B	30.2	30.0	30.4	30.0	28.8	29.9	−0.7
Grok 4.20	27.4	28.0	32.0	33.4	32.6	30.7	5.3
Best response	0	0	0	0	0	0	0
<i>Always 2</i>							
Claude Haiku 4.5	20.4	21.8	20.0	21.6	21.8	21.1	0.6
Gemini 3.5 Flash-Lite	20.0	20.0	20.0	20.0	20.0	20.0	0.0
GPT-5.6 Luna	20.0	20.0	20.0	20.0	20.2	20.0	0.1
Qwen3-235B	18.0	18.0	18.6	18.0	18.0	18.1	0.0
Grok 4.20	30.8	32.0	28.0	28.8	30.0	29.9	−2.0
Best response	0	0	0 or 20	20	20	8 or 12	20
<i>Always 4</i>							
Claude Haiku 4.5	15.8	15.0	17.0	15.0	13.8	15.3	−1.0
Gemini 3.5 Flash-Lite	11.4	11.6	9.8	11.2	10.6	10.9	−0.6
GPT-5.6 Luna	4.0	4.2	4.8	5.0	4.6	4.5	0.7
Qwen3-235B	21.8	20.8	19.4	20.8	19.2	20.4	−1.3
Grok 4.20	33.0	34.2	31.2	37.0	29.2	32.9	−0.5
Best response	0	0	0	0	0	0	0
<i>Copy the group</i>							
Claude Haiku 4.5	21.2	21.0	20.0	22.4	21.0	21.1	0.6
Gemini 3.5 Flash-Lite	20.0	20.0	20.0	20.0	20.0	20.0	0.0
GPT-5.6 Luna	20.0	20.2	19.8	19.8	20.0	20.0	−0.2
Qwen3-235B	18.0	18.0	18.6	18.0	18.0	18.1	0.0
Grok 4.20	30.0	28.0	30.0	30.0	32.0	30.0	2.0
Best response	0	0	0 or 20	20	20	8 or 12	20

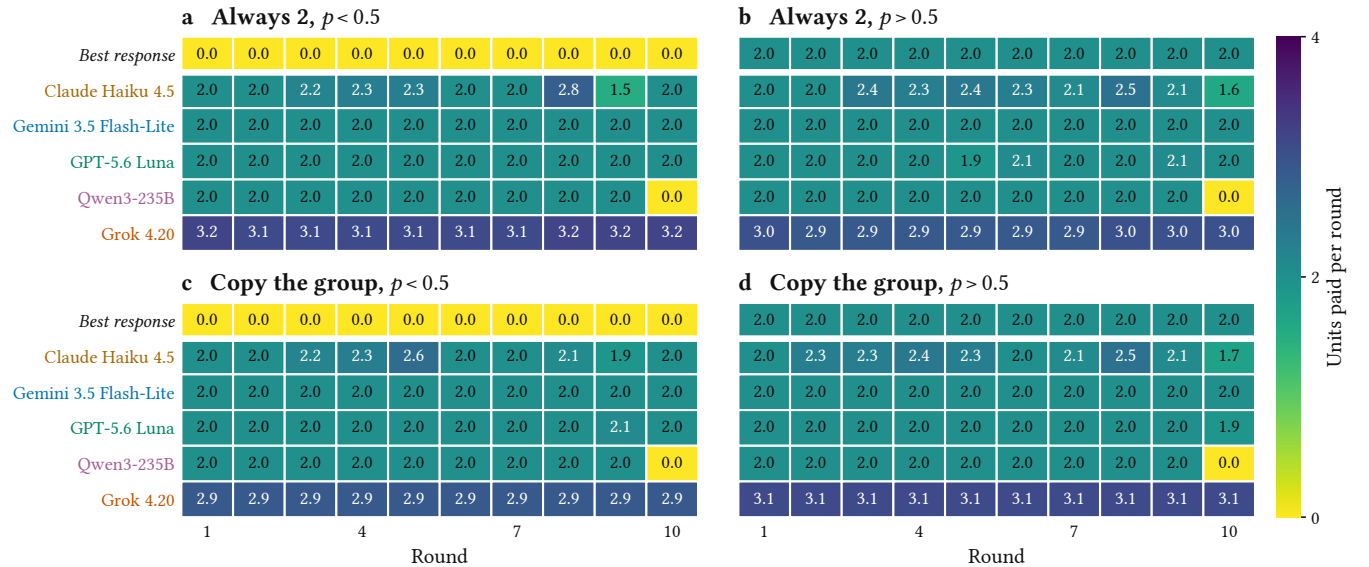


Figure S5: Mean payment in each round beside partners who always pay 2 (top) and partners who copy the group (bottom), at low risk ($p = 0.1$ and 0.3 , left) and high risk ($p = 0.7$ and 0.9 , right); 20 games per row. The top row of each panel is the best response. Above p^* any sequence that adds up to 20 is a best response, and the row shows their round-by-round average, which is 2.

S9.3 Outcomes and distance from the best response

Table S21 turns payments into outcomes. Beside partners who always pay 4, every model reaches the target in every game, so the whole gap to the best response is money paid for nothing; it ranges from small for GPT-5.6 Luna to most of the endowment for Grok 4.20. Beside fair-share partners, Claude Haiku 4.5, Gemini 3.5 Flash-Lite and GPT-5.6 Luna reach the target in almost every game and are exactly optimal in many games at or above the pivot, while Qwen3-235B almost never reaches the target.

Table S21: Outcomes by kind of partner, pooled over five risk levels (50 games per model). *Expected payoff* is the model seat's; *Gap to best response* is the best-response payoff minus it; *Exactly optimal* counts games with no gap. For the best-response rows beside fair-share partners, the target count is 20 or 30 depending on how the tie at p^* is broken.

	Reached	Expected	Gap to best response		Exactly optimal
Model	(of 50)	payoff	Mean	95% CI	(of 50)
<i>Always 0</i>					
Claude Haiku 4.5	0	16.3	3.7	[3.1, 4.3]	0
Gemini 3.5 Flash-Lite	0	13.5	6.5	[5.5, 7.6]	0
GPT-5.6 Luna	0	16.3	3.7	[3.1, 4.3]	0
Qwen3-235B	0	4.9	15.1	[12.6, 17.5]	0
Grok 4.20	0	5.3	14.7	[12.5, 17.0]	0
Best response	0	20.0	0.0		50
<i>Always 2</i>					
Claude Haiku 4.5	48	18.4	6.4	[4.6, 8.1]	17
Gemini 3.5 Flash-Lite	50	20.0	4.8	[3.0, 6.6]	30
GPT-5.6 Luna	49	19.7	5.1	[3.4, 6.9]	27
Qwen3-235B	3	11.5	13.3	[12.0, 14.4]	3
Grok 4.20	50	10.1	14.7	[11.4, 18.1]	16
Best response	20 or 30	24.8	0.0		50
<i>Always 4</i>					
Claude Haiku 4.5	50	24.7	15.3	[13.8, 16.9]	0
Gemini 3.5 Flash-Lite	50	29.1	10.9	[10.4, 11.4]	0
GPT-5.6 Luna	50	35.5	4.5	[4.2, 4.8]	0
Qwen3-235B	50	19.6	20.4	[19.8, 21.1]	0
Grok 4.20	50	7.1	32.9	[30.7, 35.2]	0
Best response	50	40.0	0.0		50
<i>Copy the group</i>					
Claude Haiku 4.5	47	18.4	6.4	[4.6, 8.3]	18
Gemini 3.5 Flash-Lite	50	20.0	4.8	[3.0, 6.6]	30
GPT-5.6 Luna	48	19.5	5.3	[3.5, 7.1]	28
Qwen3-235B	3	11.5	13.3	[11.9, 14.4]	3
Grok 4.20	50	10.0	14.8	[11.7, 18.0]	14
Best response	20 or 30	24.8	0.0		50

Beside partners who always pay 0, the target cannot be reached, and every unit a model pays is a unit that a partner paying nothing keeps. Table S22 shows this cost at each risk level. It shrinks as the risk rises only because the lottery is more likely to destroy what the partner kept.

Table S22: Beside partners who always pay 0: expected payoff of one of these partners minus the model seat’s expected payoff (ten games per risk level, 50 over all risks). Here it equals the gap to the best response in Table S21.

Model	Risk p					All risks	
	0.1	0.3	0.5	0.7	0.9	Mean	95% CI
Claude Haiku 4.5	6.1	5.0	4.1	2.6	0.8	3.7	[3.1, 4.3]
Gemini 3.5 Flash-Lite	11.3	8.3	7.1	4.3	1.7	6.5	[5.5, 7.6]
GPT-5.6 Luna	6.1	4.2	4.5	2.6	1.2	3.7	[3.1, 4.3]
Qwen3-235B	27.2	21.0	15.2	9.0	2.9	15.1	[12.6, 17.5]
Grok 4.20	24.7	19.6	16.0	10.0	3.3	14.7	[12.5, 17.0]

S9.4 Partner versus risk

Table S23 splits the variation in what a model pays beside the three fixed-amount partners. For an exact best responder in the same games, risk and its interaction with the partner explain half of the variation; this uses a payment of 20 at the tie $p = p^*$, and breaking the tie toward 0 would raise the share further. For four models the partner kind explains most of the variation and risk almost none. Grok 4.20 is different: most of its variation is residual, which Table S24 traces to its first move.

Table S23: Share of the variation in the model seat’s total explained by the partner kind, the risk and their interaction, over partners who always pay 0, 2 or 4 (150 games per model; balanced two-way sums of squares). *Risk and interaction* is computed before rounding, so it can differ by 1 from the sum of the two rounded columns.

Model	Share of the variation in the total (%)				
	Partner kind	Risk	Partner kind $\times$ risk	Risk and interaction	Residual
Claude Haiku 4.5	69	0	1	2	29
Gemini 3.5 Flash-Lite	69	1	4	5	26
GPT-5.6 Luna	92	1	2	3	5
Qwen3-235B	92	1	1	2	6
Grok 4.20	3	2	6	8	90
Best response	50	17	33	50	0

In round 1 there is no history, so every kind of partner produces the same prompt, and the first move is sampled before the model can learn anything about its partners. Four models open with 2 in all 800 of their games. Grok 4.20 opens with 4 in about half of its games and then mostly repeats its opening, repeating a 4 in 9.4 and a 2 in 7.9 of the ten rounds on average (Table S24).

Table S24: Top: Grok 4.20’s opening move and its mean ten-round total after each opening, by kind of partner (50 games each). No model ever opens with 0. Bottom: share of the variation in each model’s total explained on its own by the opening move, by the partner kind (all four kinds) and by the risk (one-way sums of squares over 200 games). Because this uses all four kinds and one-way shares, the partner share can differ slightly from Table S23.

	Games opening with			Mean total after opening with		
	0	2	4	0	2	4
Grok 4.20 beside						
Always 0	0	24	26	28.1		33.1
Always 2	0	24	26	20.0		39.1
Always 4	0	24	26	25.3		40.0
Copy the group	0	25	25	20.0		40.0
All four kinds	0	97	103	23.3		38.0
Model	Share of the variation in the total explained by (%)					
	Opening move		Partner kind		Risk	
Claude Haiku 4.5	0		72		0	
Gemini 3.5 Flash-Lite	0		76		1	
GPT-5.6 Luna	0		94		1	
Qwen3-235B	0		93		0	
Grok 4.20	74		2		1	

S9.5 Paying after the outcome is settled

Table S25 separates what a model pays before the history settles the outcome from what it pays afterwards, beside partners who always pay 0 or always pay 4. Payment before settlement can reflect a model that does not yet know its partners. Payment afterwards cannot help under any belief about the partners. Gemini 3.5 Flash-Lite, Claude Haiku 4.5 and GPT-5.6 Luna stop almost completely once the outcome is settled, and GPT-5.6 Luna never pays after settlement. Qwen3-235B and Grok 4.20 keep paying in almost every game.

Table S25: Top: after which round the outcome is settled, over all 250 games with each kind of partner. Bottom: units the model seat pays per game before and after settlement (50 games per entry), the interval for the units paid after, and the number of games with no payment after settlement.

Always 0					Always 4			
<i>Games settled after each round, all five models (of 250)</i>								
Round 5	0				26			
Round 6	236				224			
Round 7	14				0			
Model	Units paid		After:	No waste	Units paid		After:	No waste
	Before	After	95% CI	(of 50)	Before	After	95% CI	(of 50)
Claude Haiku 4.5	7.7	0.1	[0.0, 0.2]	49	14.2	1.1	[0.7, 1.6]	33
Gemini 3.5 Flash-Lite	13.6	0.2	[0.0, 0.6]	47	10.9	0.0	[0.0, 0.0]	50
GPT-5.6 Luna	8.5	0.0	[0.0, 0.0]	50	4.5	0.0	[0.0, 0.0]	50
Qwen3-235B	17.8	12.1	[12.0, 12.2]	0	12.3	8.1	[7.7, 8.5]	0
Grok 4.20	19.6	11.1	[9.8, 12.4]	1	16.7	16.2	[15.0, 17.4]	0

S9.6 Fair-share partners

Table S26 shows the most common sequence of moves beside the two kinds of fair-share partner. Gemini 3.5 Flash-Lite plays the exact split in every game. Qwen3-235B’s most common sequence pays 2 for nine rounds and 0 in the last, which leaves the table exactly one payment of 2 short of the target, and this sequence accounts for almost all of its games. Grok 4.20 splits between the exact split and paying 4 in every round.

Table S26: Beside partners who always pay 2 (top) or copy the group (bottom), pooled over five risk levels (50 games per model): games reaching the target, mean total, the most common sequence of the model’s own moves and the number of games that play it, and, among games that miss the target, the most common shortfall and how many games have it. Empty cells: no game missed.

	Reached	Mean	Most common sequence		Most common shortfall	
Model	(of 50)	total	Own moves, rounds 1 to 10	Games	Units	Games
<i>Always 2</i>						
Claude Haiku 4.5	48	21.1	2 2 2 2 2 2 2 2 2 2	22	2	2
Gemini 3.5 Flash-Lite	50	20.0	2 2 2 2 2 2 2 2 2 2	50		
GPT-5.6 Luna	49	20.0	2 2 2 2 2 2 2 2 2 2	46	2	1
Qwen3-235B	3	18.1	2 2 2 2 2 2 2 2 2 0	47	2	47
Grok 4.20	50	29.9	2 2 2 2 2 2 2 2 2 2 or 4 4 4 4 4 4 4 4 4 4	24 each		
<i>Copy the group</i>						
Claude Haiku 4.5	47	21.1	2 2 2 2 2 2 2 2 2 2	21	2	3
Gemini 3.5 Flash-Lite	50	20.0	2 2 2 2 2 2 2 2 2 2	50		
GPT-5.6 Luna	48	20.0	2 2 2 2 2 2 2 2 2 2	46	2	2
Qwen3-235B	3	18.1	2 2 2 2 2 2 2 2 2 0	47	2	47
Grok 4.20	50	30.0	2 2 2 2 2 2 2 2 2 2 or 4 4 4 4 4 4 4 4 4 4	25 each		

S10 Two models at one table

This section supports Section 4.2.2 of the paper and its Figures 3 and 4. Every pair of models shares a table, with one to five seats for the first-named model, at $p = 0.1, 0.5$ and 0.9 , with ten games per cell: 150 games per pair and 1,500 in all. “Alone” means self-play of the same model at the same three risk levels, pooling the base grid and the rerun, which use the same prompt (20 games per model and risk level).

S10.1 Numbers behind the main text

Statement in the paper, made precise	Value
Largest payoff gap per seat between two models at one table; it is Grok 4.20 against GPT-5.6 Luna	14.5
Pair with the lowest target rate; games reaching the target (%)	Gemini 3.5 Flash-Lite and Qwen3-235B; 9
Round 10 with the pool at 108: decisions paying 0 (%), Qwen3-235B; Gemini 3.5 Flash-Lite	92; 0.1
Target reached, of 30: five Gemini 3.5 Flash-Lite seats and one Qwen3-235B seat	11
Gemini 3.5 Flash-Lite alone (%)	100
Target reached, of 30: five GPT-5.6 Luna seats and one Qwen3-235B seat	9
GPT-5.6 Luna alone (%)	78
Target reached, of 30: five Claude Haiku 4.5 seats and one Qwen3-235B seat; the same with Grok 4.20	30; 30
Missed tables containing Qwen3-235B; of which reaching 120 after replacing all its round-10 zeros by 2, other actions fixed	323; 298 (92%)
Target reached, of 30: five Qwen3-235B seats and one Grok 4.20 seat; and one Claude Haiku 4.5 seat	25; 7
GPT-5.6 Luna: units per seat beside five Grok 4.20 seats; beside five seats of another model	10.5; 19.1
Units per seat beside five Grok 4.20 seats: Gemini 3.5 Flash-Lite; Claude Haiku 4.5	14.6; 16.1
Qwen3-235B: units per seat in all tables with Grok 4.20; in all tables with other models	20.8; 19.2
Share of the shortfall below 2 per round that comes after the pool reaches 120, beside Grok 4.20 (%):	91; 88; 32; 72
Claude Haiku 4.5; Gemini 3.5 Flash-Lite; GPT-5.6 Luna; Qwen3-235B	

S10.2 Who earns more at a shared table

Table S27 gives the numbers behind the upper triangle of Figure 3 of the paper. By Proposition 2, the seat that pays less earns more at the same table, so the payoff gap mirrors the payment gap. Every model earns clearly more than Grok 4.20, and no interval of those gaps comes near zero. Among the other four models the gaps are small; Qwen3-235B is slightly ahead of all three, because it pays slightly less.

Table S27: Two models at one table (150 games per pair, pooled over seat splits and risk levels). Top: expected payoff per seat of the row model minus that of the column model in the same games, with its interval below. Bottom: units paid per seat, row minus column.

Row model	Claude Haiku 4.5	Gemini 3.5 Flash-Lite	GPT-5.6 Luna	Qwen3 235B	Grok 4.20
<i>Expected payoff per seat, row model minus column model</i>					
Claude Haiku 4.5	<i>self</i>	−1.9 [−2.2, −1.5]	−2.7 [−3.1, −2.3]	−0.6 [−1.1, −0.1]	+9.7 [8.4, 10.9]
Gemini 3.5 Flash-Lite	+1.9 [1.5, 2.2]	<i>self</i>	−0.1 [−0.1, 0.0]	−0.8 [−0.9, −0.6]	+11.3 [9.8, 12.8]
GPT-5.6 Luna	+2.7 [2.3, 3.1]	+0.1 [0.0, 0.1]	<i>self</i>	−0.2 [−0.4, 0.0]	+14.5 [12.8, 16.4]
Qwen3-235B	+0.6 [0.1, 1.1]	+0.8 [0.6, 0.9]	+0.2 [0.0, 0.4]	<i>self</i>	+8.2 [7.3, 9.0]
Grok 4.20	−9.7 [−10.9, −8.4]	−11.3 [−12.8, −9.8]	−14.5 [−16.4, −12.8]	−8.2 [−9.0, −7.3]	<i>self</i>
<i>Units paid per seat, row model minus column model</i>					
Claude Haiku 4.5	<i>self</i>	+1.9	+3.0	+0.9	−9.7
Gemini 3.5 Flash-Lite	−1.9	<i>self</i>	+0.1	+1.5	−11.3
GPT-5.6 Luna	−3.0	−0.1	<i>self</i>	+0.7	−14.8
Qwen3-235B	−0.9	−1.5	−0.7	<i>self</i>	−8.4
Grok 4.20	+9.7	+11.3	+14.8	+8.4	<i>self</i>

S10.3 Which tables reach the target

Table S28 gives the numbers behind the lower triangle of Figure 3. Table S29 splits each pair by how many seats the first model holds and by risk level. Tables that mix Qwen3-235B with Gemini 3.5 Flash-Lite or GPT-5.6 Luna fail at almost every split, and only improve when Qwen3-235B holds a single seat. Tables with Grok 4.20 reach the target in the great majority of games, with two exceptions: five GPT-5.6 Luna seats beside one Grok 4.20 seat, and four or five Qwen3-235B seats beside Grok 4.20. Success barely changes with the risk level, as the self-play results would predict.

Table S28: Games reaching the target: count out of 150 for each pair, and out of 60 for self-play on the diagonal (the base grid and the rerun at $p = 0.1, 0.5$ and 0.9).

Model	Claude Haiku 4.5	Gemini 3.5 Flash-Lite	GPT-5.6 Luna	Qwen3 235B	Grok 4.20
Claude Haiku 4.5	60/60	149/150	110/150	109/150	150/150
Gemini 3.5 Flash-Lite	149/150	60/60	137/150	13/150	150/150
GPT-5.6 Luna	110/150	137/150	47/60	15/150	142/150
Qwen3-235B	109/150	13/150	15/150	0/60	140/150
Grok 4.20	150/150	150/150	142/150	140/150	60/60

Table S29: Games reaching the target (%). (a) By the number of seats the first-named model holds, pooled over risk levels (30 games per entry; *All*: 150). (b) By risk level, pooled over seat splits (50 games per entry), with self-play at the same risk levels (20 games per entry).

(a) By seats held by the first model							
First model	Second model	Seats held by the first model					All
		1	2	3	4	5	
Claude Haiku 4.5	Gemini 3.5 Flash-Lite	100	100	97	100	100	99
Claude Haiku 4.5	GPT-5.6 Luna	53	60	73	83	97	73
Claude Haiku 4.5	Qwen3-235B	23	70	77	93	100	73
Claude Haiku 4.5	Grok 4.20	100	100	100	100	100	100
Gemini 3.5 Flash-Lite	GPT-5.6 Luna	90	87	93	87	100	91
Gemini 3.5 Flash-Lite	Qwen3-235B	0	0	7	0	37	9
Gemini 3.5 Flash-Lite	Grok 4.20	100	100	100	100	100	100
GPT-5.6 Luna	Qwen3-235B	3	7	3	7	30	10
GPT-5.6 Luna	Grok 4.20	97	100	100	100	77	95
Qwen3-235B	Grok 4.20	100	100	100	83	83	93
All pairs		67	72	75	75	82	74

(b) By risk level					
First model	Second model	Risk p			
		0.1	0.5	0.9	
Claude Haiku 4.5	Gemini 3.5 Flash-Lite	98	100	100	
Claude Haiku 4.5	GPT-5.6 Luna	66	72	82	
Claude Haiku 4.5	Qwen3-235B	66	74	78	
Claude Haiku 4.5	Grok 4.20	100	100	100	
Gemini 3.5 Flash-Lite	GPT-5.6 Luna	86	90	98	
Gemini 3.5 Flash-Lite	Qwen3-235B	8	8	10	
Gemini 3.5 Flash-Lite	Grok 4.20	100	100	100	
GPT-5.6 Luna	Qwen3-235B	10	8	12	
GPT-5.6 Luna	Grok 4.20	98	92	94	
Qwen3-235B	Grok 4.20	94	94	92	
<i>Self-play</i>					
Claude Haiku 4.5		100	100	100	
Gemini 3.5 Flash-Lite		100	100	100	
GPT-5.6 Luna		80	75	80	
Qwen3-235B		0	0	0	
Grok 4.20		100	100	100	

S10.4 The last round

Before round 10, a pool of exactly 108 means that the fair share from every seat completes the target, and that no single seat can do it alone. Table S30 counts what each model pays in that situation, over all mixed tables. Qwen3-235B pays nothing in almost every such decision, while the other models rarely do. Replacing all Qwen3-235B round-10 zeros by 2 would bring 298 of the 323 failed tables containing it to the target. This arithmetic repair holds every other recorded action fixed; it does not show that changing one seat suffices or predict responses to an actual intervention.

Table S30: Top: round-10 decisions at mixed tables whose pool after round 9 is exactly 108, by model: share paying 0, number of decisions and number of games. Bottom: missed mixed tables, those with a Qwen3-235B seat, and those among them that reach 120 when Qwen3-235B’s round-10 zeros are replaced by 2.

Model	Paid 0 (%)	Decisions	Games
Claude Haiku 4.5	8.7	149	74
Gemini 3.5 Flash-Lite	0.1	956	287
GPT-5.6 Luna	1.4	775	245
Qwen3-235B	92.3	598	198
Grok 4.20	0.0	78	48
Two-model tables that missed the target			385
with a Qwen3-235B seat			323
reaching 120 if Qwen3-235B paid 2 instead of 0 in round 10			298 (92%)

S10.5 Slack and generous partners

Table S31 gives the numbers behind Figure 4 of the paper and extends it. Part (a) shows that one Qwen3-235B seat breaks tables of Gemini 3.5 Flash-Lite and GPT-5.6 Luna but not tables of Claude Haiku 4.5 or Grok 4.20, and, in the last column, that only Grok 4.20 pays enough to carry five Qwen3-235B seats to the target most of the time. Part (b) shows that the more Grok 4.20 seats a table holds, the less Claude Haiku 4.5, Gemini 3.5 Flash-Lite and GPT-5.6 Luna pay, while Qwen3-235B pays slightly more. Part (c) shows that this under-matching happens only beside Grok 4.20: beside five seats of any other model, each model pays close to the fair share.

Table S32 asks when the payment falls short beside Grok 4.20. The shortfall of a seat in a round is how far its payment falls below 2. Claude Haiku 4.5, Gemini 3.5 Flash-Lite and Qwen3-235B fall short mostly after the pool has reached 120, which fits stopping once the outcome is settled. GPT-5.6 Luna falls short mostly before, so it pays less from the start of the game.

Table S31: Slack at mixed tables, pooled over risk levels, with 95% intervals. (a) Games reaching the target: six seats of the model (60 games), five seats and one Qwen3-235B seat (30 games), and one seat beside five Qwen3-235B seats (30 games). (b) Units paid per seat of the model beside one to five Grok 4.20 seats (30 games each), and beside five seats of another model (90 games). (c) Units paid by a single seat beside five seats of each other model (30 games each).

<i>(a) Target reached</i>						
Model	Six seats of the model		Five seats and one Qwen3-235B seat			One seat and five Qwen3-235B seats
	%	95% CI	Reached	%	95% CI	
Claude Haiku 4.5	100	[100, 100]	30/30	100	[100, 100]	7/30
Gemini 3.5 Flash-Lite	100	[100, 100]	11/30	37	[20, 53]	0/30
GPT-5.6 Luna	78	[68, 88]	9/30	30	[13, 47]	1/30
Grok 4.20	100	[100, 100]	30/30	100	[100, 100]	25/30

<i>(b) Units paid per seat of the model, by the seats of the other model</i>						
Model	Grok 4.20 seats					Five seats of another model
	1	2	3	4	5	
Claude Haiku 4.5	20.9 [20.5, 21.2]	19.5 [18.8, 20.2]	19.4 [19.0, 19.8]	19.0 [18.1, 20.0]	16.1 [14.9, 17.1]	21.2 [20.8, 21.6]
Gemini 3.5 Flash-Lite	19.5 [19.2, 19.9]	19.1 [18.7, 19.4]	17.9 [17.0, 18.8]	16.3 [15.4, 17.1]	14.6 [13.1, 16.0]	19.8 [19.6, 19.9]
GPT-5.6 Luna	18.7 [18.2, 19.3]	18.0 [17.4, 18.6]	14.3 [12.6, 15.9]	12.0 [10.1, 14.0]	10.5 [8.5, 12.7]	19.1 [18.8, 19.5]
Qwen3-235B	19.1 [18.9, 19.3]	20.6 [19.9, 21.3]	20.2 [19.6, 20.7]	23.4 [22.5, 24.3]	20.8 [20.2, 21.4]	19.7 [19.2, 20.2]

<i>(c) Units paid by one seat of the model beside five seats of each other model</i>						
Model	Claude Haiku 4.5	Gemini 3.5 Flash-Lite	GPT-5.6 Luna	Qwen3 235B	Grok 4.20	
Claude Haiku 4.5	<i>self</i>	20.9 [20.5, 21.5]	21.4 [20.6, 22.2]	21.4 [20.7, 22.2]	16.1 [14.9, 17.1]	
Gemini 3.5 Flash-Lite	19.2 [18.9, 19.5]	<i>self</i>	20.1 [20.0, 20.3]	20.0 [20.0, 20.0]	14.6 [13.1, 16.0]	
GPT-5.6 Luna	18.1 [17.1, 18.9]	20.1 [20.0, 20.2]	<i>self</i>	19.3 [18.9, 19.6]	10.5 [8.5, 12.7]	
Qwen3-235B	20.5 [19.4, 21.7]	18.9 [18.5, 19.3]	19.7 [19.0, 20.6]	<i>self</i>	20.8 [20.2, 21.4]	

Table S32: Shortfall below 2 per seat and round beside one to five Grok 4.20 seats (150 games per model, pooled over seat splits and risk levels), before and after the round in which the pool reaches 120. *Per round*: mean over games of the mean shortfall per seat and round; *Games*: games with at least one round in that phase. Right: units short per seat per game before and after, and the share that comes after.

Model	Before the pool reached 120			After the pool reached 120			Units short per seat per game		
	Per round	95% CI	Games	Per round	95% CI	Games	Before	After	After (%)
Claude Haiku 4.5	0.03	[0.02, 0.05]	150	1.88	[1.83, 1.93]	124	0.3	2.7	91
Gemini 3.5 Flash-Lite	0.04	[0.02, 0.06]	150	1.83	[1.73, 1.92]	88	0.3	2.3	88
GPT-5.6 Luna	0.46	[0.38, 0.53]	150	2.00	[2.00, 2.00]	74	3.8	1.8	32
Qwen3-235B	0.05	[0.04, 0.06]	150	1.25	[1.12, 1.38]	113	0.5	1.3	72

S11 Selection among models

This section supports Section 4.3 of the paper and its Figure 5. It gives the method in full, the payoffs that feed it, and every selection setting we computed.

S11.1 Method

Payoff functions. For two models A and B and a risk level p , $\pi_A(k)$ is the mean over games of the mean expected payoff of the A seats at a table with k seats of A and $6 - k$ seats of B . For $1 \leq k \leq 5$ it comes from the mixed tables (ten games per value). For $k = 6$ it is A 's self-play payoff at the same risk level (20 games, pooling the base grid and the rerun). Tables S33 to S35 list every value.

Fitness. A population holds N agents, i of type A and $N - i$ of type B , and tables of six are drawn at random without replacement. An A agent sits with $k - 1$ other A agents among its five tablemates with the hypergeometric probability $\mathcal{H}(k - 1; N - 1, i - 1)$, and a B agent sits with k A agents with probability $\mathcal{H}(k; N - 1, i)$, where

$$\mathcal{H}(k; M, m) = \binom{m}{k} \binom{M - m}{5 - k} / \binom{M}{5}.$$

The fitness of each type is its payoff averaged over these tables, as in Equation (2) of the paper. With $N = 6$ the whole population is a single table, so the A agents earn exactly $\pi_A(i)$ and the B agents $\pi_B(6 - i)$.

Fixation. Under the pairwise comparison rule, an agent copies another with probability $1/(1 + e^{-\beta \Delta f})$, where Δf is the other agent's fitness minus its own. The probability that one A newcomer takes over a population of B is

$$\rho_{B \rightarrow A} = \left(1 + \sum_{j=1}^{N-1} \prod_{i=1}^j e^{-\beta (f_A(i) - f_B(i))} \right)^{-1}.$$

We write the transition from a population of B to a population of A as $B \rightarrow A$, the direction of the arrows in Figure S7. A takeover is favoured by selection when $\rho_{B \rightarrow A} > 1/N$, the probability for a neutral newcomer.

Rare-mutation Fermi selection. In the limit where a new model enters rarely, the population almost always holds a single model. From a population of one model, a newcomer of each of the four other models arrives with equal probability and takes over with the probability above. These transitions form a Markov chain over the five one-model populations, and its stationary distribution is the *stationary mass* of each model. The population's target rate and payoff per seat are the self-play target rates and payoffs weighted by these masses. The masses in Figure 5 of the paper and the invasion diagrams in Figure S7 are computed with EGTtools, and the analysis code checks that EGTtools and its own implementation give the same fixation probabilities and masses to within 10^{-7} . For the two strongest-selection settings at $N = 60$ and $N = 120$ at $p = 0.1$, the eigenvector method sees two unit eigenvalues because some transition probabilities are too small to represent; an exact computation in log space gives the same masses. We report this finite-intensity, frequency-dependent process, not an infinite-intensity ranking or group reproduction. Changing N changes both sampling probabilities and the fixation path.

Gradient of selection. For a population with i agents of type A out of N , the expected change in the share $x = i/N$ of type A in one step of the pairwise comparison rule is

$$G(x) = x(1 - x) \frac{N}{N - 1} \tanh\left(\frac{\beta}{2} (f_A(i) - f_B(i))\right).$$

Where $G > 0$ selection pushes the share of A up, and where $G < 0$ it pushes it down. A zero of G at which the sign changes from positive to negative is stable, and one at which it changes from negative to positive is unstable. This population-level gradient is distinct from the table-composition payoffs plotted in Figure 5 of the paper.

Uncertainty. Each of 1,000 bootstrap replicates resamples the games of every mixed cell and every self-play cell, rebuilds all payoff functions and repeats the whole analysis. We report how often the winner of the point estimate stays on top, and percentile intervals for the population’s target rate and payoff.

S11.2 Numbers behind the main text

Statement in the paper, made precise	Value
One table ($N = 6$, $\beta = 1$): Qwen3-235B’s mass at $p = 0.1$; 0.5; 0.9	0.89; 0.66; 0.47
One table: population target rate at $p = 0.5$; 0.9 (%)	28; 47
a model picked at random (%)	75; 76
stronger selection, $\beta = 10$ (%)	0; 1
Larger population ($N = 30$, $\beta = 1$): Gemini 3.5 Flash-Lite’s mass at $p = 0.5$; 0.9	1.00; 1.00
Larger population at $p = 0.9$: population target rate (%); payoff per seat	100; 20.0
Larger population at $p = 0.1$: model with the largest mass	Qwen3-235B
Winners unchanged for $\beta = 10$, for N of 60 and 120, and with grid-only self-play	yes
Smallest share of bootstrap replicates in which the winner stays on top (%): pooled self-play; grid only	93; 90
Weak selection ($\beta = 0.1$), one table ($N = 6$): largest mass of any model at any risk level	0.34
model with the largest mass at $p = 0.5$; 0.9	GPT-5.6 Luna; GPT-5.6 Luna
Weak selection, larger population ($N = 30$): model with the largest mass at $p = 0.5$; 0.9	Gemini 3.5 Flash-Lite; Gemini 3.5 Flash-Lite
Larger population: smallest population from which Gemini 3.5 Flash-Lite stays the winner at $p = 0.9$; at $p = 0.5$	7; 8
Cells (pair, split, risk) in which a single seat beside five Grok 4.20 seats earns more than they do	12 of 12
Cells with Qwen3-235B seats; Qwen3-235B significantly below its tablemates; significantly above	60; one; 40

S11.3 Payoff functions

Tables S33 to S35 give $\pi_X(k)$ for every ordered pair of models. Reading along a row shows how a model’s payoff changes as its own share of the table grows. Two patterns drive the selection results. Beside Qwen3-235B, Gemini 3.5 Flash-Lite and GPT-5.6 Luna earn little whenever the target is missed, which is at almost every split. And Grok 4.20 earns less than its tablemates, because it pays the most.

Table S33: Payoff functions at $p = 0.1$: mean expected payoff per seat of the model in the first column at a table with k of its seats beside $6 - k$ seats of the model in the second column (ten games per entry for $k \leq 5$; $k = 6$ is self-play, 20 games, and is the same for every partner).

Model	Beside	Seats of the model at the table					
		1	2	3	4	5	6
Claude Haiku 4.5	Gemini 3.5 Flash-Lite	19.00	18.30	18.31	18.55	18.20	18.97
	GPT-5.6 Luna	18.02	17.17	17.47	17.99	18.40	18.97
	Qwen3-235B	17.08	18.00	18.00	18.15	18.72	18.97
	Grok 4.20	23.00	21.50	20.27	20.10	19.56	18.97
Gemini 3.5 Flash-Lite	Claude Haiku 4.5	20.40	21.20	20.27	20.00	20.12	19.98
	GPT-5.6 Luna	19.60	19.60	19.60	19.65	19.96	19.98
	Qwen3-235B	18.00	18.09	18.20	18.00	18.71	19.98
	Grok 4.20	21.60	22.00	21.60	20.85	20.84	19.98
GPT-5.6 Luna	Claude Haiku 4.5	22.20	20.67	20.83	20.54	19.16	19.70
	Gemini 3.5 Flash-Lite	20.00	19.69	19.86	19.87	19.67	19.70
	Qwen3-235B	19.08	19.39	18.25	18.49	18.62	19.70
	Grok 4.20	28.38	25.30	21.60	22.75	21.16	19.70
Qwen3-235B	Claude Haiku 4.5	17.20	17.89	20.29	19.09	19.48	19.49
	Gemini 3.5 Flash-Lite	19.32	19.80	19.52	19.62	19.55	19.49
	GPT-5.6 Luna	18.92	19.42	19.76	18.64	19.19	19.49
	Grok 4.20	20.00	18.20	20.67	19.70	20.64	19.49
Grok 4.20	Claude Haiku 4.5	14.60	12.00	14.33	10.45	10.60	6.68
	Gemini 3.5 Flash-Lite	11.20	14.60	14.00	12.00	14.20	6.68
	GPT-5.6 Luna	12.00	12.70	16.40	12.55	10.60	6.68
	Qwen3-235B	13.80	13.01	13.73	11.90	10.88	6.68

Table S34: Payoff functions at $p = 0.5$, as in Table S33.

Model	Beside	Seats of the model at the table					
		1	2	3	4	5	6
Claude Haiku 4.5	Gemini 3.5 Flash-Lite	19.40	18.30	17.73	18.60	18.56	18.75
	GPT-5.6 Luna	14.90	14.05	13.97	17.33	17.38	18.75
	Qwen3-235B	11.80	17.50	15.50	18.55	18.76	18.75
	Grok 4.20	24.40	21.00	20.93	19.90	18.92	18.75
Gemini 3.5 Flash-Lite	Claude Haiku 4.5	20.80	20.50	20.53	19.80	20.04	20.00
	GPT-5.6 Luna	18.90	18.95	19.00	18.00	20.00	20.00
	Qwen3-235B	10.00	10.00	10.00	10.00	13.96	20.00
	Grok 4.20	25.00	23.70	21.13	20.75	20.36	20.00
GPT-5.6 Luna	Claude Haiku 4.5	21.40	20.60	17.33	16.90	16.14	17.53
	Gemini 3.5 Flash-Lite	20.00	18.10	19.07	19.00	19.04	17.53
	Qwen3-235B	10.50	10.15	10.43	10.35	14.06	17.53
	Grok 4.20	29.20	29.30	26.80	21.20	16.72	17.53
Qwen3-235B	Claude Haiku 4.5	20.60	15.85	17.70	16.23	13.82	10.79
	Gemini 3.5 Flash-Lite	14.40	10.85	10.87	10.98	10.58	10.79
	GPT-5.6 Luna	13.30	10.60	10.67	10.90	10.66	10.79
	Grok 4.20	19.00	16.10	18.80	17.00	19.72	10.79
Grok 4.20	Claude Haiku 4.5	14.80	12.70	13.20	11.45	7.12	5.88
	Gemini 3.5 Flash-Lite	15.20	13.10	14.67	8.35	8.24	5.88
	GPT-5.6 Luna	11.50	13.80	9.20	6.80	8.40	5.88
	Qwen3-235B	13.40	10.60	10.07	13.35	4.68	5.88

Table S35: Payoff functions at $p = 0.9$, as in Table S33.

Model	Beside	Seats of the model at the table					
		1	2	3	4	5	6
Claude Haiku 4.5	Gemini 3.5 Flash-Lite	18.80	18.60	17.60	18.50	19.00	18.85
	GPT-5.6 Luna	11.18	10.05	18.07	18.20	18.60	18.85
	Qwen3-235B	4.72	15.50	16.40	19.10	18.64	18.85
	Grok 4.20	24.40	20.40	20.60	21.55	18.88	18.85
Gemini 3.5 Flash-Lite	Claude Haiku 4.5	21.20	20.30	20.33	19.85	19.92	20.00
	GPT-5.6 Luna	19.80	18.10	19.93	19.95	20.00	20.00
	Qwen3-235B	2.00	2.01	3.81	2.00	9.16	20.00
	Grok 4.20	29.60	25.50	23.53	21.25	20.20	20.00
GPT-5.6 Luna	Claude Haiku 4.5	21.00	21.20	20.73	11.41	12.95	16.42
	Gemini 3.5 Flash-Lite	19.80	20.00	20.00	18.21	19.92	16.42
	Qwen3-235B	3.80	3.66	2.03	3.71	7.47	16.42
	Grok 4.20	30.60	29.40	28.73	21.95	15.62	16.42
Qwen3-235B	Claude Haiku 4.5	20.80	17.70	18.62	14.68	5.85	2.15
	Gemini 3.5 Flash-Lite	9.52	2.20	3.73	2.17	2.18	2.15
	GPT-5.6 Luna	7.26	3.78	2.15	4.00	3.96	2.15
	Grok 4.20	18.60	15.50	20.07	18.85	13.46	2.15
Grok 4.20	Claude Haiku 4.5	14.80	7.80	10.87	13.15	2.08	2.62
	Gemini 3.5 Flash-Lite	17.20	12.00	7.40	5.50	1.24	2.62
	GPT-5.6 Luna	7.40	11.70	5.47	6.20	5.80	2.62
	Qwen3-235B	8.78	6.20	11.67	8.15	3.48	2.62

S11.4 Every selection setting

The winners of Figure 5 of the paper are robust across several checks: each winner remains on top in at least 93% of bootstrap samples, and the same winners emerge at $\beta = 10$, at N of 60 and 120 and when using grid-only self-play payoffs. Under weak selection ($\beta = 0.1$), the largest one-table mass is only 0.34; GPT-5.6 Luna narrowly leads at moderate and high risk. Which model wins therefore depends on the base-prompt payoffs, whose signatures vary with wording.

Table S36 lists the stationary mass of each model, and the population’s target rate and payoff, for one table ($N = 6$) and for larger populations ($N = 30, 60$ and 120), at the selection intensities $\beta = 0.1, 1$ and 10 ; the two largest populations are computed only for $\beta \geq 1$. Figure S6 shows how the masses move with the selection intensity over a wider range. In a population of one table, stronger selection concentrates the weight on Qwen3-235B at every risk level and group success falls to almost zero. In the larger population, Gemini 3.5 Flash-Lite already leads at moderate and high risk under weak selection and takes all the weight from $\beta = 1$ on, while Qwen3-235B takes it at low risk. Grok 4.20 loses its weight under both population sizes as soon as selection is more than very weak.

Table S36: Stationary mass of each model under each selection setting, with the population’s self-play target rate and expected payoff per seat. *Random model:* every model equally weighted.

Population	N	β	Claude Haiku 4.5	Gemini 3.5 Flash-Lite	GPT-5.6 Luna	Qwen3 235B	Grok 4.20	Target (%)	Payoff per seat
$p = 0.1$									
One table	6	0.1	0.12	0.24	0.31	0.34	0.00	60	19.6
	6	1	0.00	0.01	0.09	0.89	0.00	9	19.5
	6	10	0.00	0.00	0.00	1.00	0.00	0	19.5
Many tables	30	0.1	0.01	0.08	0.22	0.69	0.00	27	19.6
	30	1	0.00	0.00	0.00	1.00	0.00	0	19.5
	30	10	0.00	0.00	0.00	1.00	0.00	0	19.5
	60	1	0.00	0.00	0.00	1.00	0.00	0	19.5
	60	10	0.00	0.00	0.00	1.00	0.00	0	19.5
	120	1	0.00	0.00	0.00	1.00	0.00	0	19.5
	120	10	0.00	0.00	0.00	1.00	0.00	0	19.5
Random model			0.20	0.20	0.20	0.20	0.20	76	17.0
$p = 0.5$									
One table	6	0.1	0.12	0.26	0.33	0.29	0.00	63	16.4
	6	1	0.01	0.07	0.26	0.66	0.00	28	13.3
	6	10	0.00	0.00	0.00	1.00	0.00	0	10.8
Many tables	30	0.1	0.01	0.66	0.32	0.02	0.00	90	19.0
	30	1	0.00	1.00	0.00	0.00	0.00	100	20.0
	30	10	0.00	1.00	0.00	0.00	0.00	100	20.0
	60	1	0.00	1.00	0.00	0.00	0.00	100	20.0
	60	10	0.00	1.00	0.00	0.00	0.00	100	20.0
	120	1	0.00	1.00	0.00	0.00	0.00	100	20.0
	120	10	0.00	1.00	0.00	0.00	0.00	100	20.0
Random model			0.20	0.20	0.20	0.20	0.20	75	14.6
$p = 0.9$									
One table	6	0.1	0.13	0.29	0.31	0.27	0.00	67	13.9
	6	1	0.00	0.23	0.29	0.47	0.00	47	10.6
	6	10	0.00	0.00	0.01	0.99	0.00	1	2.2
Many tables	30	0.1	0.00	0.79	0.21	0.00	0.00	96	19.3
	30	1	0.00	1.00	0.00	0.00	0.00	100	20.0
	30	10	0.00	1.00	0.00	0.00	0.00	100	20.0
	60	1	0.00	1.00	0.00	0.00	0.00	100	20.0
	60	10	0.00	1.00	0.00	0.00	0.00	100	20.0
	120	1	0.00	1.00	0.00	0.00	0.00	100	20.0
	120	10	0.00	1.00	0.00	0.00	0.00	100	20.0
Random model			0.20	0.20	0.20	0.20	0.20	76	12.0

Table S37 gives the bootstrap check for the two settings of the paper, with self-play from the grid and the rerun together and from the grid alone. The winners are the same in both cases and stay on top in the large majority of replicates. Some

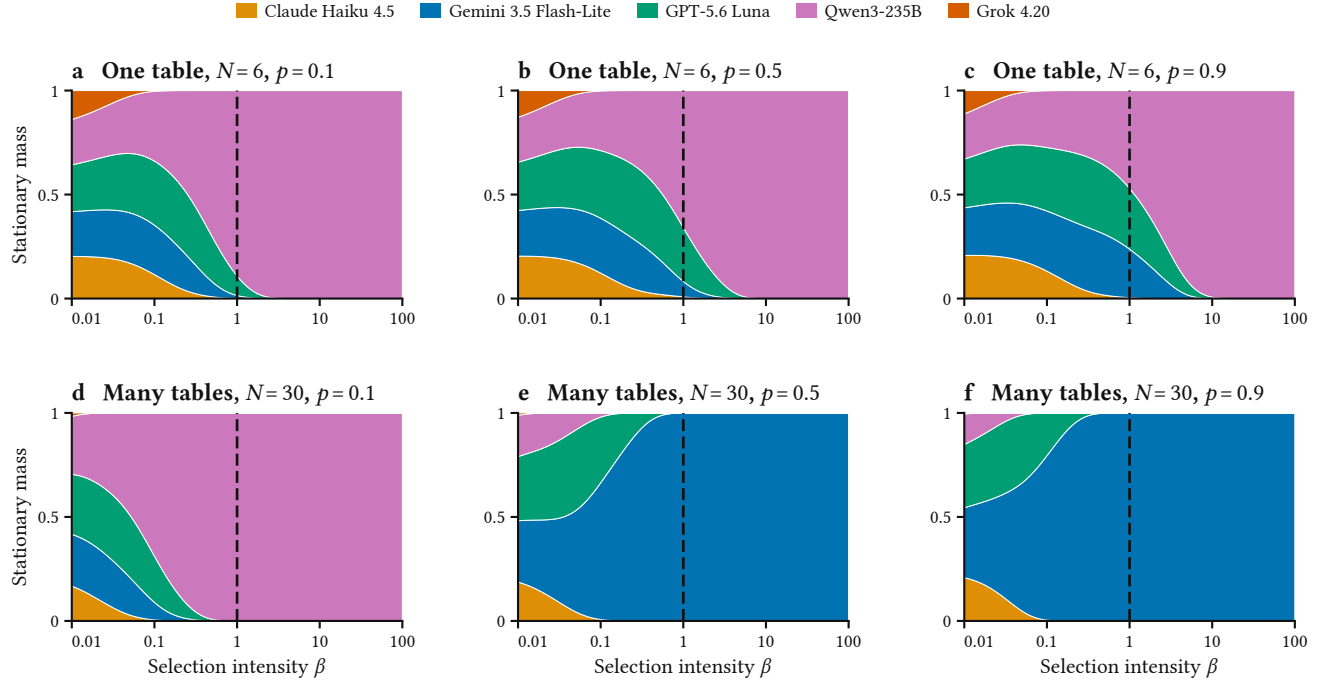


Figure S6: Stationary mass of the five models as the selection intensity β grows from 0.01 to 100. Top: a population of one table ($N = 6$). Bottom: a larger population ($N = 30$). The dashed line marks $\beta = 1$, the setting of Figure 5 of the paper.

intervals for the target rate are wide, because a population's target rate depends on which model wins and the models differ sharply in self-play success.

Figure 5 of the paper shows the stationary masses and the table-composition payoffs for one pair. Figure S7 gives the full invasion diagrams behind those masses, for every pair of models at the three risk levels and both population sizes. Each arrow marks a takeover that selection favours.

Table S37: Bootstrap check of the two settings of Figure 5 ($\beta = 1$; one table, $N = 6$; larger population, $N = 30$), over 1,000 replicates: the winner of the point estimate, the share of replicates in which it stays on top, and the population’s target rate and payoff per seat with 95% intervals.

Population	p	Top model	Top in (%)	Target (%)	95% CI	Payoff per seat	95% CI
<i>Self-play from the grid and the rerun (20 games per model and risk level)</i>							
One table	0.1	Qwen3-235B	98	9	[2, 42]	19.5	[19.4, 19.6]
	0.5	Qwen3-235B	95	28	[11, 51]	13.3	[11.8, 15.4]
	0.9	Qwen3-235B	93	47	[37, 61]	10.6	[8.8, 13.0]
Many tables	0.1	Qwen3-235B	95	0	[0, 67]	19.5	[19.4, 19.7]
	0.5	Gemini 3.5 Flash-Lite	99	100	[98, 100]	20.0	[19.8, 20.0]
	0.9	Gemini 3.5 Flash-Lite	99	100	[100, 100]	20.0	[20.0, 20.0]
<i>Self-play from the grid only (10 games per model and risk level)</i>							
One table	0.1	Qwen3-235B	97	9	[1, 47]	19.5	[19.4, 19.6]
	0.5	Qwen3-235B	95	29	[12, 53]	13.3	[11.9, 15.6]
	0.9	Qwen3-235B	94	44	[33, 59]	10.0	[8.0, 12.6]
Many tables	0.1	Qwen3-235B	94	0	[0, 72]	19.5	[19.4, 19.7]
	0.5	Gemini 3.5 Flash-Lite	90	100	[97, 100]	20.0	[19.7, 20.0]
	0.9	Gemini 3.5 Flash-Lite	98	100	[100, 100]	20.0	[20.0, 20.0]

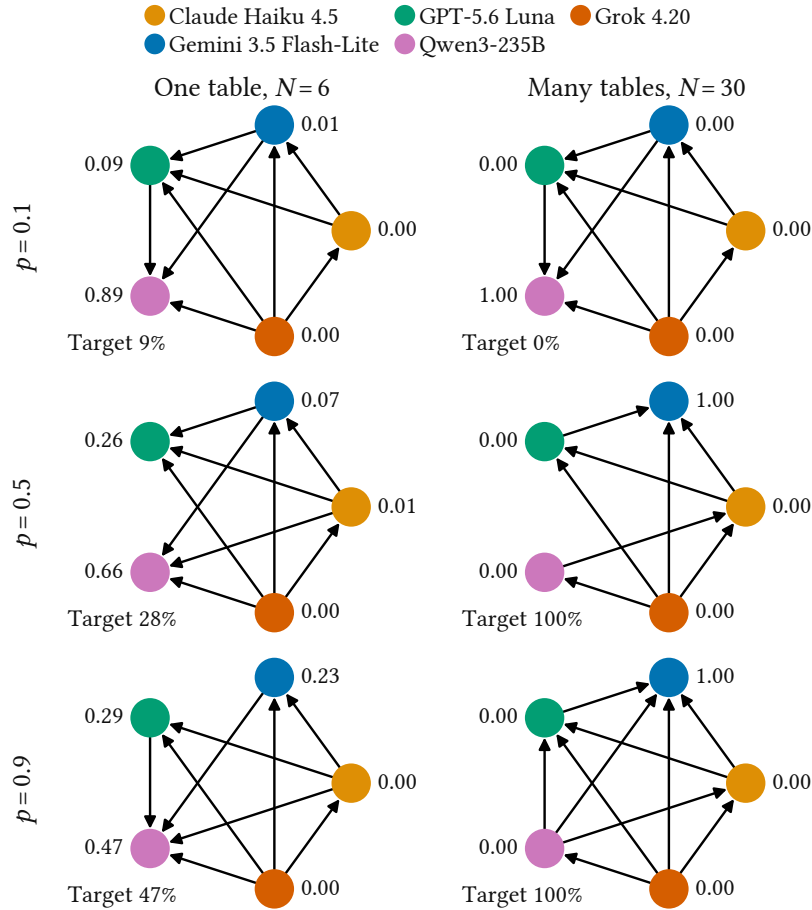


Figure S7: Invasion diagrams (rare-mutation Fermi selection, $\beta = 1$), computed and drawn with EGTtools. A node is a one-model population, with its stationary mass beside it. An arrow from A to B means that one B newcomer takes over an A population with more than the neutral probability $1/N$. Left: one table, $N = 6$. Right: larger randomly mixed population, $N = 30$. Bottom left: self-play target rate weighted by mass.

S11.5 Facts about single tables

Table S38 checks, cell by cell, the two table-level facts that the selection results rest on. A single seat of any model beside five Grok 4.20 seats earns more than they do in every cell. Qwen3-235B is significantly ahead of its tablemates in most cells and significantly behind in one, which is about what chance allows among so many tests. Part (c) shows how large the population must be before selection favours Gemini 3.5 Flash-Lite: a population only slightly larger than one table is enough.

Table S38: (a) Expected payoff of a single seat of each model beside five Grok 4.20 seats (ten games per risk level). (b) Qwen3-235B's expected payoff per seat against its tablemates in each cell of ten games, exact one-sided sign-flip tests in both directions. (c) The winner as the population size N grows from 6 to 120.

<i>(a) One seat beside five Grok 4.20 seats: expected payoff per seat</i>				
Model	p	One seat	Grok 4.20 seats	Ahead
Claude Haiku 4.5	0.1	23.00	10.60	One seat
	0.5	24.40	7.12	One seat
	0.9	24.40	2.08	One seat
Gemini 3.5 Flash-Lite	0.1	21.60	14.20	One seat
	0.5	25.00	8.24	One seat
	0.9	29.60	1.24	One seat
GPT-5.6 Luna	0.1	28.38	10.60	One seat
	0.5	29.20	8.40	One seat
	0.9	30.60	5.80	One seat
Qwen3-235B	0.1	20.00	10.88	One seat
	0.5	19.00	4.68	One seat
	0.9	18.60	3.48	One seat
<i>(b) Qwen3-235B against its tablemates, exact one-sided sign-flip test per cell, $\alpha = 0.05$</i>				
Cells (pair, seats, p) with Qwen3-235B seats				60
Qwen3-235B significantly below its tablemates				1
Qwen3-235B significantly above its tablemates				40
Below: $p = 0.5$, two Qwen3-235B seats beside four Claude Haiku 4.5 seats, payoff gap -2.70 , one-sided $P = 0.004$				
<i>(c) Winner by population size, $\beta = 1$, N from 6 to 120</i>				
p	Winner at smaller N		From N	Winner up to $N = 120$
0.1	none		6	Qwen3-235B
0.5	Qwen3-235B ($N = 6$ to 7)		8	Gemini 3.5 Flash-Lite
0.9	Qwen3-235B ($N = 6$)		7	Gemini 3.5 Flash-Lite

S11.6 Single entrants, candidate sets and welfare

The winners depend on the candidate set: without Qwen3-235B, GPT-5.6 Luna wins the one-table population at all tested risks, and without Gemini 3.5 Flash-Lite, it wins the larger population at moderate and high risk.

The following post-hoc checks reuse the observed payoffs without new model calls. We compare each single entrant’s payoff with both resident self-play and its actual tablemates, then compute fixation along the full composition path. These are different comparisons: a favourable initial payoff gap alone does not determine fixation or stationary mass. We also remove each model in turn and rebuild the mutation chain over the retained strategies. This checks dependence on the candidate set, not on the prompt. Every risk and both population sizes are reported, with no new uncertainty intervals.

At $p = 0.5$, reaching the target is not required for optimal welfare: zero contribution also earns 20 per seat in expectation. We therefore report stationary-mass-weighted expected payoff and its share of the welfare optimum as well as target rate. The outcomes describe one-model populations in the rare-mutation limit, not the success of mixed groups during invasion.

Post-hoc selection diagnostics. All entries are point estimates; no new intervals are computed. Names: Haiku = Claude Haiku 4.5; Flash-Lite = Gemini 3.5 Flash-Lite; Luna = GPT-5.6 Luna; Qwen = Qwen3-235B; Grok = Grok 4.20.

(A1) Single entrant and five residents, $p = 0.1$; fixation at $\beta = 1$.

Resident B	Entrant A	u_B^{self}	$u_A(1)$	$u_B(5)$	Δ_{self}	Δ_{table}	ρ_6	ρ_{30}
Haiku	Flash-Lite	18.97	20.40	18.20	1.43	2.20	0.89	0.8
Haiku	Luna	18.97	22.20	18.40	3.23	3.80	0.98	0.96
Haiku	Qwen	18.97	17.20	18.72	-1.77	-1.52	0.081	0.00012
Haiku	Grok	18.97	14.60	19.56	-4.37	-4.96	3.7e-19	3.1e-114
Flash-Lite	Haiku	19.98	19.00	20.12	-0.98	-1.12	5.8e-05	1.6e-23
Flash-Lite	Luna	19.98	20.00	19.96	0.02	0.04	0.22	0.048
Flash-Lite	Qwen	19.98	19.32	18.71	-0.66	0.61	0.6	0.12
Flash-Lite	Grok	19.98	11.20	20.84	-8.78	-9.64	1.7e-18	1.3e-112
Luna	Haiku	19.70	18.02	19.16	-1.68	-1.14	5.8e-07	3.6e-33
Luna	Flash-Lite	19.70	19.60	19.67	-0.10	-0.07	0.11	0.0048
Luna	Qwen	19.70	18.92	18.62	-0.78	0.30	0.43	0.062
Luna	Grok	19.70	12.00	21.16	-7.70	-9.16	1.4e-24	1.2e-142
Qwen	Haiku	19.48	17.08	19.48	-2.40	-2.40	0.0015	3.2e-14
Qwen	Flash-Lite	19.48	18.00	19.55	-1.48	-1.55	0.00066	1.4e-15
Qwen	Luna	19.48	19.08	19.19	-0.40	-0.11	0.052	4.1e-06
Qwen	Grok	19.48	13.80	20.64	-5.68	-6.84	2.6e-16	7.1e-101
Grok	Haiku	6.68	23.00	10.60	16.32	12.40	1	1
Grok	Flash-Lite	6.68	21.60	14.20	14.92	7.40	1	1
Grok	Luna	6.68	28.38	10.60	21.70	17.78	1	1
Grok	Qwen	6.68	20.00	10.88	13.32	9.12	1	1

(A2) Single entrant and five residents, $p = 0.5$; fixation at $\beta = 1$.

Resident B	Entrant A	u_B^{self}	$u_A(1)$	$u_B(5)$	Δ_{self}	Δ_{table}	ρ_6	ρ_{30}
Haiku	Flash-Lite	18.75	20.80	18.56	2.05	2.24	0.89	0.87
Haiku	Luna	18.75	21.40	17.38	2.65	4.02	0.98	0.94
Haiku	Qwen	18.75	20.60	18.76	1.85	1.84	0.21	2.6e-10
Haiku	Grok	18.75	14.80	18.92	-3.95	-4.12	1.2e-20	1e-122
Flash-Lite	Haiku	20.00	19.40	20.04	-0.60	-0.64	0.0001	1.1e-22
Flash-Lite	Luna	20.00	20.00	20.00	0.00	0.00	0.19	0.00016
Flash-Lite	Qwen	20.00	14.40	13.96	-5.60	0.44	0.48	9.4e-14
Flash-Lite	Grok	20.00	15.20	20.36	-4.80	-5.16	4.8e-23	7.8e-137
Luna	Haiku	17.52	14.90	16.14	-2.62	-1.24	3.8e-07	1.2e-30
Luna	Flash-Lite	17.52	18.90	19.04	1.38	-0.14	0.13	0.55
Luna	Qwen	17.52	13.30	14.06	-4.22	-0.76	0.14	8.1e-14
Luna	Grok	17.52	11.50	16.72	-6.02	-5.22	1.2e-32	1.3e-180
Qwen	Haiku	10.79	11.80	13.82	1.01	-2.02	0.026	0.52
Qwen	Flash-Lite	10.79	10.00	10.58	-0.79	-0.58	0.012	8.1e-06
Qwen	Luna	10.79	10.50	10.66	-0.29	-0.16	0.073	0.0017
Qwen	Grok	10.79	13.40	19.72	2.61	-6.32	1.9e-17	6.6e-93
Grok	Haiku	5.88	24.40	7.12	18.52	17.28	1	1
Grok	Flash-Lite	5.88	25.00	8.24	19.12	16.76	1	1
Grok	Luna	5.88	29.20	8.40	23.32	20.80	1	1
Grok	Qwen	5.88	19.00	4.68	13.12	14.32	1	1

(A3) Single entrant and five residents, $p = 0.9$; fixation at $\beta = 1$.

Resident B	Entrant A	u_B^{self}	$u_A(1)$	$u_B(5)$	Δ_{self}	Δ_{table}	ρ_6	ρ_{30}
Haiku	Flash-Lite	18.85	21.20	19.00	2.35	2.20	0.88	0.9
Haiku	Luna	18.85	21.00	18.60	2.15	2.40	0.91	0.89
Haiku	Qwen	18.85	20.80	18.64	1.95	2.16	0.56	1.1e-22
Haiku	Grok	18.85	14.80	18.88	-4.05	-4.08	1.5e-25	5.4e-153
Flash-Lite	Haiku	20.00	18.80	19.92	-1.20	-1.12	9.8e-05	1.5e-22
Flash-Lite	Luna	20.00	19.80	20.00	-0.20	-0.20	0.16	1.1e-06
Flash-Lite	Qwen	20.00	9.52	9.16	-10.48	0.36	0.26	4.1e-31
Flash-Lite	Grok	20.00	17.20	20.20	-2.80	-3.00	4.7e-34	1.4e-197
Luna	Haiku	16.41	11.18	12.95	-5.23	-1.77	1.2e-05	7.3e-21
Luna	Flash-Lite	16.41	19.80	19.92	3.39	-0.12	0.14	0.93
Luna	Qwen	16.41	7.26	7.47	-9.15	-0.21	0.17	1e-24
Luna	Grok	16.41	7.40	15.62	-9.01	-8.22	1.1e-39	1.5e-219
Qwen	Haiku	2.15	4.72	5.85	2.57	-1.13	0.021	0.87
Qwen	Flash-Lite	2.15	2.00	2.18	-0.15	-0.18	0.11	0.071
Qwen	Luna	2.15	3.80	3.96	1.65	-0.16	0.11	0.53
Qwen	Grok	2.15	8.78	13.46	6.63	-4.68	1.2e-21	1.4e-104
Grok	Haiku	2.62	24.40	2.08	21.78	22.32	1	1
Grok	Flash-Lite	2.62	29.60	1.24	26.98	28.36	1	1
Grok	Luna	2.62	30.60	5.80	27.98	24.80	1	1
Grok	Qwen	2.62	18.60	3.48	15.98	15.12	1	1

$\Delta_{\text{self}} = u_A(1) - u_B^{\text{self}}$ compares separate conditions; $\Delta_{\text{table}} = u_A(1) - u_B(5)$ compares seats in the same mixed games. Mixed entries average ten games; resident self-play averages twenty. ρ_N is fixation probability from one entrant in a population of size N , using all table compositions and $\beta = 1$, not just the first payoff gap. Neutral fixation is $1/N$. These comparisons do not establish evolutionary stability.

(B1) All candidate sets, $N = 6$ and $\beta = 1$.

Omitted model	$p = 0.1$: winner (mass)	$p = 0.5$: winner (mass)	$p = 0.9$: winner (mass)
None (all five)	Qwen (0.891)	Qwen (0.658)	Qwen (0.470)
Haiku	Qwen (0.894)	Qwen (0.709)	Qwen (0.490)
Flash-Lite	Qwen (0.887)	Qwen (0.586)	Qwen (0.588)
Luna	Qwen (0.995)	Qwen (0.916)	Qwen (0.665)
Qwen	Luna (0.665)*	Luna (0.588)*	Luna (0.535)*
Grok	Qwen (0.891)	Qwen (0.658)	Qwen (0.470)

(B2) All candidate sets, $N = 30$ and $\beta = 1$.

Omitted model	$p = 0.1$: winner (mass)	$p = 0.5$: winner (mass)	$p = 0.9$: winner (mass)
None (all five)	Qwen (1.000)	Flash-Lite (1.000)	Flash-Lite (1.000)
Haiku	Qwen (1.000)	Flash-Lite (1.000)	Flash-Lite (1.000)
Flash-Lite	Qwen (1.000)	Luna (1.000)*	Luna (1.000)*
Luna	Qwen (1.000)	Flash-Lite (1.000)	Flash-Lite (1.000)
Qwen	Luna (0.909)*	Flash-Lite (1.000)	Flash-Lite (1.000)
Grok	Qwen (1.000)	Flash-Lite (1.000)	Flash-Lite (1.000)

*Winner differs from the full five-model set, including when that winner is omitted. Removing a candidate leaves every retained payoff unchanged and changes only the available models and mutation destinations.

(C) Target rate and welfare at $p = 0.5$, $\beta = 1$; opt = 20.

Omitted model	N	Target (%)	Welfare/seat	Welfare/opt (%)	Uniform target (%)	Uniform welfare/opt (%)
None	6	27.7	13.28	66.4	75.0	72.9
None	30	100.0	20.00	100.0	75.0	72.9
Haiku	6	23.3	12.90	64.5	68.8	67.8
Haiku	30	100.0	20.00	100.0	68.8	67.8
Flash-Lite	6	31.4	13.60	68.0	68.8	66.2
Flash-Lite	30	75.0	17.52	87.6	68.8	66.2
Luna	6	8.4	11.54	57.7	75.0	69.3
Luna	30	100.0	20.00	100.0	75.0	69.3
Qwen	6	85.3	18.54	92.7	93.8	77.7
Qwen	30	100.0	20.00	100.0	93.8	77.7
Grok	6	27.7	13.28	66.4	68.8	83.8
Grok	30	100.0	20.00	100.0	68.8	83.8

Outcomes weight each retained model's self-play by its stationary mass; uniform assigns equal weight to retained models. These are rare-mutation monomorphic outcomes, not success rates during invasion. At $p = 0.5$, zero contribution and an exact target both attain welfare 20, although their target rates are 0 and 100%. Low target rate alone therefore does not imply welfare loss.

S12 Strategies, the last round and equilibrium benchmarks

This section reports how the models play the last round, and supports three statements in Sections 4.1.1 and 4.2.2 of the paper: how the models compare with the equilibria of Proposition 1, how each model’s moves can be described by a simple rule, and how partners respond to Grok 4.20’s opening move. All tables use the games played with the base prompt.

S12.1 The last round

The last move reveals three rules. Qwen3-235B’s final-round skipping appears to be driven by the timing of the move rather than by the state of the collective pool. Across all base-prompt games, it skips the last contribution as often when its contribution is needed as when the others’ fair shares already suffice (93% of seats in both cases), and it almost never stops earlier once the target is met. Human contributions similarly decline over the rounds of repeated public goods games (Andreoni, 1988). GPT-5.6 Luna skips in 2% of cases when the target needs its contribution but in 53% when the others’ fair shares would suffice. When six copies each count on another seat to contribute, tables ahead of the fair pace of 12 per round often miss the target, while tables exactly on pace reach it. Claude Haiku 4.5 stops contributing in the final round mostly when the target has already been reached, showing that models can infer the current pool size from the prompt.

Table S39 splits the last move of every seat by the state of the pool before round 10. In self-play, GPT-5.6 Luna tables that reach round 10 ahead of the fair pace reach the target in 27% of 26 games, against 92% of 99 games exactly on pace: when every seat reasons that another will pay, too many skip.

Table S39: Share of seats paying nothing in round 10, by the state of the pool after round 9, and in rounds 7 to 9 once the pool has reached 120, with 95% bootstrap intervals over games in the table file.

Model	Target already reached		Others’ 2s suffice		Own 2 is pivotal		Rounds 7 to 9, reached	
	Paid 0 (%)	Games	Paid 0 (%)	Games	Paid 0 (%)	Games	Paid 0 (%)	Games
Claude Haiku 4.5	92 [90, 94]	337	13 [11, 14]	413	9 [5, 13]	135	90 [87, 93]	106
Gemini 3.5 Flash-Lite	94 [91, 97]	164	9 [7, 11]	197	0 [0, 0]	525	96 [93, 99]	97
GPT-5.6 Luna	100 [100, 100]	143	53 [50, 57]	285	2 [1, 2]	441	100 [100, 100]	83
Qwen3-235B	80 [76, 84]	221	93 [92, 95]	307	93 [92, 95]	361	1 [0, 2]	98
Grok 4.20	2 [1, 3]	633	4 [2, 6]	159	0 [0, 0]	98	0 [0, 0]	341
<i>GPT-5.6 Luna self-play: target reached (%) by the pool after round 9</i>								
Ahead of pace (110 to 118)	27 [12, 42]	26						
On pace (exactly 108)	92 [86, 97]	99						

Top: share of decisions paying 0 with a 95% bootstrap interval over games, and the number of games. Round 10 is split by the pool the seats saw before deciding: at least 120 (reached); 110 to 118, so the other five paying 2 already reach 120 (others’ 2s suffice); exactly 108 (own 2 is pivotal). The last pair of columns pools rounds 7 to 9 that start with the pool already at 120 or more. Base-prompt games: self-play, mixed tables and scripted partners.

S12.2 Play against the equilibria of Proposition 1

For $p < 1$ the worst Nash equilibrium is that nobody pays, which leaves $(1 - p)$ 40 per seat; every target-reaching equilibrium leaves 20. Table S40 counts the self-play games whose expected payoff per seat falls below the worst equilibrium. Above the pivot this happens in most games of Qwen3-235B, which misses the target after paying, and of Grok 4.20, which overpays, and never for Gemini 3.5 Flash-Lite or Claude Haiku 4.5.

Table S40: Self-play games on the risk grid whose expected payoff per seat is below that of the worst Nash equilibrium.

Model	$0.5 < p < 1$		$p < 0.5$	
	Games	Below (%)	Games	Below (%)
Claude Haiku 4.5	40	0 [0, 0]	50	100 [100, 100]
Gemini 3.5 Flash-Lite	40	0 [0, 0]	50	100 [100, 100]
GPT-5.6 Luna	40	25 [13, 38]	50	80 [68, 90]
Qwen3-235B	40	100 [100, 100]	50	100 [100, 100]
Grok 4.20	40	93 [83, 100]	50	100 [100, 100]

Share of self-play games on the risk grid whose expected payoff per seat falls below that of the worst Nash equilibrium, with a 95% bootstrap interval over games. By Proposition 1, for $p < 1$ the equilibria are $x = 0$, worth $(1 - p)40$ per seat, and profiles with $X = T$ and $x_i \leq 40p$, which exist only for $p \geq 0.5$ and are worth 20 per seat, never less than $(1 - p)40$. The worst equilibrium is therefore nobody pays, $(1 - p)40$, at every $p < 1$; below $p = 0.5$ it is also the only one. $p = 1$ is left out because no payoff can fall below 0 there, and $p = 0.5$ because the two kinds of equilibrium tie.

S12.3 Simple rules describe the moves

Table S41 compares each model’s moves with named rules and with a small decision tree fitted by cross-validation. For every model one rule predicts most decisions, and the tree adds little. Removing the risk level from the tree’s inputs does not lower its accuracy for any model, which is the sense in which the paper calls the play a default. A rule close to the risk-neutral optimum predicts far fewer moves than the best rule for every model.

Table S41: How well named rules and a fitted tree predict each model’s moves.

Accuracy (%)	Claude Haiku 4.5	Gemini 3.5 Flash-Lite	GPT-5.6 Luna	Qwen3 235B	Grok 4.20
<i>Named rules (best per model in bold)</i>					
Always 2	81 [80, 82]	96 [95, 97]	89 [87, 90]	86 [85, 86]	41 [39, 43]
Repeat own move	78 [77, 78]	98 [98, 98]	94 [93, 94]	88 [87, 88]	86 [85, 86]
Copy the group	81 [80, 81]	95 [94, 96]	91 [90, 92]	84 [84, 85]	68 [67, 70]
Last-round skipper	81 [81, 82]	89 [88, 89]	85 [84, 86]	94 [94, 95]	41 [38, 43]
Stop when settled	86 [86, 86]	99 [98, 99]	92 [91, 93]	87 [86, 87]	41 [39, 43]
Pool tracker	85 [84, 86]	99 [99, 99]	92 [91, 93]	87 [86, 87]	41 [39, 44]
Risk-neutral step	57 [54, 60]	63 [60, 67]	63 [60, 67]	58 [55, 61]	23 [21, 25]
<i>Baselines and fitted policy</i>					
Majority class	81 [80, 82]	96 [95, 97]	89 [87, 90]	86 [85, 86]	58 [56, 61]
Majority move	2	2	2	2	4
Tree, cross-validated	87 [86, 87]	99 [99, 99]	95 [95, 96]	96 [96, 97]	91 [91, 91]
Tree without risk p	87 [86, 87]	99 [99, 99]	95 [95, 96]	96 [96, 97]	91 [91, 91]
Change without p (pp)	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.1 [0.0, 0.2]
Tree minus best rule (pp)	0.5 [0.3, 0.7]	0.1 [0.0, 0.2]	1.9 [1.7, 2.2]	2.3 [1.9, 2.7]	5.2 [4.8, 5.7]
Games	940	940	940	940	940
Decisions	28,400	28,400	28,400	28,400	28,400

Accuracy is the share of a model’s decisions a policy predicts exactly, with a 95% bootstrap interval over games; base-prompt games (self-play, mixed tables, scripted partners). Always 2. Repeat own move: last round’s own move, 2 in round 1. Copy the group: the others’ mean last round rounded to 0, 2 or 4, 2 in round 1. Last-round skipper: 2, then 0 in round 10. Stop when settled: 2 until the target is reached or out of reach, then 0. Pool tracker: 0 once settled, else the share each seat still needs per round rounded up to 2 or 4. Risk-neutral step: 0 at $p < 0.5$; at $p \geq 0.5$, 2 until the target is reached, then 0. Tree: depth 5, at least 50 decisions per leaf, five folds that keep each game in one fold; features are the own previous move, the others’ previous mean, the pool against the fair pace, rounds left, a last-round flag, reached, out of reach, the amount each seat still needs per round, and p . Changes are paired over the same decisions.

S12.4 Responses to Grok 4.20’s opening move

Grok 4.20 chooses its first move before any history exists, and that move varies across repetitions of a cell. Table S42 relates a partner’s payments to Grok 4.20’s mean opening move within each cell. Higher openings are associated with lower payments by GPT-5.6 Luna while the pool is short. Claude Haiku 4.5’s payment in that phase barely changes and Gemini 3.5 Flash-Lite’s changes only slightly; both pay less in total mainly through earlier target completion. Qwen3-235B pays more. This is an observational association, not a randomised opening intervention or proof of an internal reasoning mechanism.

Table S42: Change in a partner seat’s payment per unit of Grok 4.20’s mean opening move, within cells of the two-model tables, with 95% bootstrap intervals over games.

Partner model	Games	Partner’s payment per seat		
		Total, rounds 2 to 10	Per round, pool below 120	Per round, pool at 120 or more
Claude Haiku 4.5	150	−1.81 [−2.40, −1.40]	0.00 [−0.05, 0.06]	−0.09 [−0.22, 0.03] (124)
Gemini 3.5 Flash-Lite	150	−2.00 [−2.44, −1.61]	−0.03 [−0.05, −0.01]	−0.19 [−0.49, 0.08] (88)
GPT-5.6 Luna	150	−3.69 [−4.34, −3.05]	−0.37 [−0.45, −0.30]	0.00 [0.00, 0.00] (74)
Qwen3-235B	150	0.78 [0.19, 1.39]	0.03 [−0.06, 0.11]	0.97 [0.44, 1.52] (113)

Change in the partner’s payment per seat for one more unit in the mean opening move of the Grok 4.20 seats, which is made before any history. Mixed tables of Grok 4.20 with one other model; OLS within cells of the same number of Grok seats and the same p , controlling for the partner’s own opening move; 95% bootstrap interval resampling games within cells. Per-round columns average rounds 2 to 10 whose starting pool is below 120, or at 120 or more; the last column counts only games with such rounds (in brackets after the interval: games).